Under consideration for publication in J. Fluid Mech.

A predictive latent representation renders wake structure linearly readable in vortex-gust airfoil interactions at Re = 5000

Alberto Solera-Rico[1,2], Arnau Miró[3,4], Oriol Lehmkuhl[4] and Carlos Sanmiguel Vila[1,2]

[1]Aerial Platforms Department, Spanish National Institute for Aerospace Technology (INTA), San Martín de la Vega, 28330, Madrid, Spain

[2]Department of Aerospace Engineering, Universidad Carlos III de Madrid, Leganés, 28911, Madrid, Spain

[3]TUAREG Research Group, Universitat Politècnica de Catalunya (UPC), Carrer Colom 11, 08222 Terrassa, Spain

[4]Barcelona Supercomputing Centre (BSC), CASE Large-Scale Computational Fluid Dynamics, Plaça d'Eusebi Güell, 1-3, 08034 Barcelona, Spain

Corresponding author: Carlos Sanmiguel Vila, csanvil@inta.es

(Received xx; revised xx; accepted xx)

Extreme vortex-gust encounters challenge reduced-order models because the transient load is governed by the reorganisation of the separated wake, not only by the integrated forces. We ask which reduced state renders that wake recoverable, from the flow and from sparse sensors, at matched latent dimension. Using direct numerical simulations of a NACA 0012 airfoil at $\alpha = 14°$ and Re = 5000, perturbed by Taylor vortices spanning gust strength G, core diameter D, and wall-normal offset Y, we compare reduced states under one shared predictor and probe protocol: a joint-embedding predictive representation, a reconstructive observable-augmented autoencoder of the Fukami-Taira lineage, and a proper orthogonal decomposition basis. Probed directly from held-out fields sixteen frames after impact, the predictive latent renders the wake enstrophy linearly readable ($R^2 = 0.79$ on the in-distribution test split) where the reconstructive and linear baselines do not ($R^2 \leq 0$); the per-encounter advantage survives case-level clustering and a family-wide multiplicity correction, with the training, validation, test_b, and extrapolation test_c splits reported in the text, and a nonlinear probe narrows but does not close it, so we claim linear decodability rather than exclusive possession. Regressed on the gust parameters alone the same observable is not recovered, so the latent carries wake state beyond the parameters. The mechanism is geometric: the predictive encoding organises each encounter as a single persistent cycle whose spectrum recovers the shedding frequency as a marginally stable orbit, and its autoregressive rollout stays on the encoded manifold, whereas the reconstructive encoding fragments even the undisturbed limit cycle and its rollout leaves the manifold along directions of negligible encoded variance. The response follows the flow's own scales: the wake departure grows with gust strength and the gust resets the shedding phase. The same predictive state is the most recoverable from sparse wall pressure

($R^2 = 0.78$ at eight taps, against 0.66 and 0.34 for the alternatives). Rolled forward, the predictor tracks the encounter, which we show but do not present as a validated forecast.

Key words: vortex shedding, low-dimensional models, machine learning

1. Introduction

Small uncrewed and micro air vehicles increasingly operate in airspace where the gust velocity is comparable to or larger than their flight speed, such as urban canyons, mountainous terrain, and the wakes of buildings and ships (Jones *et al.* 2022; Mohamed *et al.* 2023). In this regime the gust ratio G, the ratio of the characteristic gust velocity to the free-stream velocity, exceeds unity, and the interaction between a discrete vortex gust and a lifting surface produces large, fast transients in the aerodynamic loads that conventional gust models do not capture (Fukami *et al.* 2024, 2025). The response is governed by the timing of the vortex impact relative to the airfoil's own shedding, by the wall-normal offset of the vortex core, and by the gust strength and core size. The load transient itself is produced by the formation, growth, and shedding of a leading-edge vortex, the dominant unsteady mechanism in these encounters (Eldredge & Jones 2019), together with the trailing-edge and wake structures it leaves behind. Even at the moderate chord Reynolds number Re = 5000, the post-stall flow at $\alpha = 14^\circ$ is massively separated, and the gust-induced disturbance interacts nonlinearly with the natural shedding cycle (Fukami *et al.* 2025); the vortical organisation of the encounter is moreover similar across laminar and turbulent regimes (Odaka *et al.* 2026), so this case is representative of the wider extreme-aerodynamic family. Resolving the full parametric envelope by simulation is expensive, so reduced-order models (ROMs) of the encounter-to-encounter and case-to-case variability are an active line of work, and a reduced model is also the object a controller would plan against.

Reduced-order models of such flows form a ladder of increasing nonlinearity. Energy-optimal linear bases, proper orthogonal decomposition (Berkooz *et al.* 1993) and dynamic mode decomposition (Schmid 2010), are compact and interpretable but limited for strongly nonlinear separation; their Koopman-operator reading (Lusch *et al.* 2018) nonetheless makes a linear latent a principled comparator rather than a strawman. Nonlinear autoencoders compress further at matched latent dimension (Murata *et al.* 2020; Lee & Carlberg 2020), and coupling an autoencoder to a learned latent dynamics forecasts better than dynamics confined to a linear subspace (Linot & Graham 2023). The extreme-aerodynamic reduced models we build on sit at the nonlinear, observable-augmented end of this ladder.

Recent ROMs for separated and extreme-aerodynamic flows, increasingly built with machine learning (Brunton *et al.* 2020), share a common ingredient: a reconstruction-trained autoencoder whose bottleneck is shaped by an auxiliary aerodynamic observable. Fukami & Taira (2023) trained a convolutional autoencoder jointly with a lift-prediction head and found that the auxiliary loss reduced the intrinsic dimension of the discovered manifold for extreme vortex-gust flows from roughly ten to three. The same observable-augmentation idea was extended across multi-source turbulent data (Fukami & Taira 2025), and was used by Solera-Rico et al. (2024) with a β-variational autoencoder and a transformer ROM on a vortex-gust airfoil dataset of the same family considered here. Across this lineage the reconstruction term anchors the latent geometry to the instantaneous flow field, and an observable-prediction term orients it along aerodynamically relevant directions. A recurring theme of the geometric-analysis literature, however, is that the latent coordinates of an autoencoder are unique only up to a diffeomorphism, so reconstruction leaves the geometry of the latent largely unconstrained, which complicates any downstream task that relies on that geometry, including dynamics, interpolation, and comparison (Tran *et al.* 2026; Kelshaw & Magri 2024).

A parallel and fast-growing line reads these flows through the information their reduced states carry rather than their energy. Transfer-entropy causality among proper-orthogonal-decomposition modes ranks which modes drive the formation of coherent structures (Martínez-Sánchez *et al.* 2023), and an informative/non-informative decomposition splits a field into the part that carries information about a target's future and a residual that carries none (Arranz & Lozano-Durán 2024). This informative-decomposition idea has very recently been brought to extreme vortex-gust airfoil interactions of the present family, extracting the field structures informative about the future lift (Koshikawa *et al.* 2026; Fukami & Araki 2026), alongside time-dependent modal bases for the same interaction (Zhong *et al.* 2025; Zamani Ashtiani & Fukami 2025) and latent-causality attributions in physical space (Cremades *et al.* 2026). Our study is complementary: rather than mapping the informative or causal modes of a single representation, we ask which encoder objective produces a latent whose forward evolution closes the full set of observables, the wake included, and we use the information lens only as a held-out diagnostic of the resulting latent (§5.1).

A complementary route to a reduced state is to drop the reconstruction term altogether and to train the encoder and a predictor end to end on latent-space forecasting, with an explicit anti-collapse regulariser in place of the decoder. This is the joint-embedding predictive architecture (JEPA) (LeCun 2022): an encoder maps the input to a representation, a predictor advances that representation in time, and the loss never reconstructs the field. Two consequences follow. The encoder is free to discard detail that is not predictive of the future, and the predictor cannot exploit pixel-level shortcuts because no field-space signal reaches it. The image and video instantiations (Assran *et al.* 2023, 2025) and the recent from-pixels variants that remove the moving-target encoder in favour of a characteristic-function regulariser (Maes *et al.* 2026; Balestriero & LeCun 2025) or a variance-covariance objective (Sobal *et al.* 2025; Bardes *et al.* 2022) have so far been evaluated on gridworld and toy visual domains. Their behaviour on a low-intrinsic-dimension physics dataset, where the data sit on a roughly five- to ten-dimensional manifold, is uncharacterised.

Reconstruction-trained nonlinear encoders compress aggressively but yield latent dynamics that are harder to forecast than an energy-optimal linear basis, a compactness-versus-forecast tradeoff documented for controlled wake flows in a companion study under review (Solera-Rico *et al.* 2025). We show this tradeoff is a consequence of the reconstruction objective: a predictive latent recovers nonlinear compactness without surrendering forecast stability. The division of labour with that companion study (Solera-Rico *et al.* 2025) is explicit: there, controlled canonical wakes and the compactness-versus-forecast tradeoff; here, the parametric vortex gust, the latent-drift mechanism we identify as the common cure, and its geometric and topological characterisation.

The model we adopt takes no gust parameters: (G, D, Y) enter neither the encoder nor the predictor. The encoder is a pure state map and the predictor advances the latent from its own history alone. The world-model framework of LeCun (2022); Ha & Schmidhuber (2018) motivates this split as an analogy and not a property we claim, since the latent dynamics function is the object a controller would plan against, and a recent gust-airfoil instance couples a physics-augmented autoencoder to a latent dynamics model (Liu *et al.* 2025). We are deliberate about scope: we do not claim an interventional or causal model (§5), and the rollout is shown but not advanced as a validated forecast. What this leaves is the property our principle turns on: a reconstruction-trained autoencoder and an energy-optimal proper orthogonal decomposition basis are not constrained to keep the dynamically relevant wake structure linearly readable in the latent, and the drift mechanism of §4.3 is the empirical consequence.

The question this paper addresses is therefore not which encoder reconstructs the flow best, but which produces a reduced state from which the dynamically relevant wake is readable. We treat this as a fluid-mechanics question with a geometric answer: a predictive objective regularised against collapse organises the latent so that the encounter is a single recurrent cycle that survives rollout, the wake is carried as a distributed code, and the autoregressive rollout stays on the encoded manifold. A reconstruction objective does not impose this, so it smooths the wake away and its rollout drifts off the manifold along the directions the decoder never constrained. The joint-embedding predictive architecture is the instrument we use to realise the principle, not the subject of the paper. We make three contributions.

First, we set up a controlled comparison in which a predictive encoder (JEPA), a reconstructive encoder in the Fukami lineage, and a POD basis are evaluated at matched latent dimension under a single shared autoregressive predictor, training recipe, and probe family, so that representation quality is attributable to the encoder alone (§3). The headline figure of merit is representational closure, whether a fixed probe reads each of six observables off the simulation-encoded latent, the wake in particular (§4.1). On held-out cases the predictive latent renders the wake linearly readable where the others do not: its representational wake-enstrophy R^2 is 0.79 where the reconstructive and linear baselines are at or below zero, and the per-encounter advantage survives the pre-registered encounter-level multiplicity correction. We phrase the result as linear decodability rather than exclusive possession, since a nonlinear probe narrows the gap. The predictor rollout points the same way, which we show but do not advance as a validated forecast.

Second, we identify the mechanism. A latent-drift diagnostic shows that the reconstructive rollout leaves its training manifold by an order of magnitude at $H = 16$ while the predictive and linear rollouts remain within it, and the departure is along directions of negligible encoded variance, which reconciles the small Euclidean drift of the reconstructive rollout with its large Mahalanobis departure (§4.3). We add a topological statement on the cleanest control: under a metric that removes coordinate scale, the predictive encoding keeps the undisturbed shedding cycle as a single persistent generator where the reconstructive encoding fragments it, so the reconstructive encoding cannot represent even the no-gust limit cycle coherently.

Third, we are explicit about what the comparison does and does not isolate. The predictive encoder differs from the reconstructive baseline not only in its objective but also in its architecture and in its auxiliary observable supervision. We therefore phrase the result as a property of the *predictive latent family*, report a model-free parametric floor that rules out the gust parameters as the explanation (§4.1), and perform matched-architecture and matched-auxiliary controls (§4.2) that attribute the wake gain to the predictive objective trained with wake-observable supervision, not to the architecture. We further show that the same predictive state is the most recoverable of the three from sparse wall pressure, so it is observable as well as a faithful carrier of the wake (§4.7), and we relate the predictor to the latent dynamics function a controller would plan against, with a closed-loop pilot reported as a limitation rather than a result (§5).

2. Flow configuration and data

2.1. *Vortex gust-airfoil interaction*

We study a NACA 0012 airfoil at angle of attack $\alpha = 14^\circ$ and chord Reynolds number $\mathrm{Re} = 5000$, perturbed by a discrete vortex gust modelled as a spanwise-oriented Taylor vortex released upstream of the leading edge. The data analysed here are our own direct numerical simulations of this configuration, computed with the GPU-enabled spectral-element solver SOD2D (Gasparino *et al.* 2024) with no subgrid-scale model, so that all scales of the separated wake are resolved at this Reynolds number. The solver is run in the low-Mach regime in which the flow is effectively incompressible. The configuration and its physical characterisation follow Fukami, Smith & Taira (2025), which is the reference for the physics, not the source of the data. The vortex has the azimuthal velocity profile

$$u_\theta(r) = u_{\theta,\max}\,\frac{r}{R}\,\exp\left[\tfrac{1}{2}\left(1 - \frac{r^2}{R^2}\right)\right], \tag{2.1}$$

which peaks at the core radius $r = R$ (Taylor 1918), and is parametrised by three scalars: the signed gust ratio $G \equiv u_{\theta,\max}/u_\infty$, the core diameter $D \equiv 2R/c$, and the wall-normal offset Y of the vortex centre, the displacement normal to the chord (the laboratory coordinates are rotated by the angle of attack to define Y). The profile and the (G, D, Y) sampling follow the configuration of Fukami *et al.* (2025). The training envelope spans $|G| \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$ with $G = 0$ as the undisturbed reference, $D \in \{0.5, 1.0, 1.5\}$ chords, and $Y \in [-0.4, +0.4]$; the value $|G| = 4$ is held out as an extrapolation boundary, for reasons that are physical and that we return to below. Convective time t is measured in chords and set to zero when the vortex centre reaches the leading edge.

Without a gust, the $\alpha = 14^\circ$ wake at this Reynolds number is massively separated and sheds periodically; in dynamical terms the undisturbed flow is a limit cycle. A gust encounter is then a transient excursion from this limit cycle and back, and it is helpful to read it in stages (Smith *et al.* 2024; Fukami *et al.* 2025; Odaka *et al.* 2026). The vortex first induces a strong wall-normal vorticity flux at the leading edge, the airfoil experiences a sharp lift excursion as a leading-edge vortex (LEV) forms, the LEV rolls up and sheds together with a trailing-edge structure as the load reaches its extremum, and the wake then relaxes back toward the baseline shedding. The sign of G sets whether the first lift excursion is positive or negative, Y sets the side and strength of the impingement, D sets how much of the encounter is dominated by the vortex core, and the impact timing sets the phase of the baseline cycle at which the disturbance arrives. The six observables defined in §2.2 are quantitative summaries of exactly this sequence.

A consequence of the source characterisation (Fukami *et al.* 2025) is central to how we read the extrapolation case. For $|G| \leq 3$ the encounter is primarily two-dimensional, so a mid-plane slice of the spanwise vorticity carries the relevant large-scale dynamics; for $|G| \geq 4$ the interaction becomes genuinely three-dimensional, with fine-scale instabilities developing during impact. We confirm this from the full three-dimensional vorticity fields: the fraction of the spanwise-vorticity (ω_z) enstrophy not captured by its spanwise mean, the content a mid-plane slice discards, is roughly constant across the training range ($|G| \leq 3$, post-impact wake median ≈ 0.20) and rises about threefold at $|G| = 4$ (≈ 0.56, over the four available cases). Because the encoder in this work consumes a single mid-plane vorticity field (§2.2), the held-out $|G| = 4$ case is not merely one parameter step beyond the training range; it is a regime in which the quantity the encoder observes no longer determines the true dynamics. We therefore treat $|G| = 4$ as a physical observability boundary rather than as a claim of parametric generalisation.

Quantity	Value	Why a referee needs it
Free-stream Mach number (or incompressible-limit confirmation)	**[PENDING:]**	Confirms the low-Mach formulation is effectively incompressible at Re = 5000.
Computational domain and spanwise extent L_z/c	**[PENDING:]**	Bounds domain blockage and confirms the span resolves the three-dimensional wake.
Element and solution-point counts	**[PENDING:]**	Fixes the spatial resolution of the spectral-element discretisation.
Minimum wall-normal spacing $\Delta n_{\min}/c$ and wall units Δn^+	**[PENDING:]**	Shows the near-wall layer is resolved without a wall model.
Time step $\Delta t\, u_\infty/c$ and maximum CFL	**[PENDING:]**	Establishes the temporal resolution and stability of the integration.
Gust-release station x_0/c	**[PENDING:]**	Sets the upstream distance over which the Taylor vortex develops before impact.
Grid and time-step sensitivity check	**[PENDING:]**	Demonstrates that the reported statistics are converged.

Table 1. Author-fill checklist of the DNS solver-resolution details required for full reproducibility (§2.2). The data are direct numerical simulations with no subgrid-scale model, so all scales are resolved and no large-eddy closure is involved; the resolution values are to be supplied by the simulation collaborators and are not yet filled in.

2.2. *Data, fields, and observables*

Each simulation case is one long run partitioned into encounters of 120 cached frames at $\Delta t = 0.05\, t/c$, one encounter per gust release. The simulations use a spectral-element discretisation of polynomial order $p = 4$, and the fields are cached on a structured grid of $192 \times 96 \times 32$ points spanning $x/c \in [-1.5, 4.5]$ and $y/c \in [-1.5, 1.5]$ with a spanwise extent of approximately one chord; this analysis subdomain matches the data-driven subdomain of the source characterisation (Fukami *et al.* 2025). The lift and drag coefficients are $C_L = F_L/(\frac{1}{2}\rho u_\infty^2 c)$ and $C_D = F_D/(\frac{1}{2}\rho u_\infty^2 c)$, obtained by integrating the pressure and viscous stresses over the airfoil surface. The remaining DNS solver-resolution details required for full reproducibility, the free-stream Mach number, the domain and spanwise extent, the element and solution-point counts, the near-wall spacing, the time step and CFL, the gust-release station, and the grid and time-step sensitivity, are listed as an author-fill checklist in table 1. The dataset comprises 84 cases partitioned into training, validation (the held-out encounters of the training cases, denoted test_a), an in-distribution test set (test_b), and an out-of-distribution test set (test_c). The in-distribution test set is stratified to span the five gust strengths, the three core diameters, and the two source groups (figure 1); test_c is the $|G| = 4$ extrapolation set. These 84 cases are separated in preprocessing into 378 encounter windows of 120 frames each, one per gust release: 226 training, 86 validation (test_a, the held-out last encounters of the training cases), 42 test_b, and 24 test_c. The undisturbed reference case is retained in training. Behind these encounter counts are far fewer distinct cases: the 42 test_b encounters come from only 10 held-out cases (five interior and five boundary tier) and the 24 test_c encounters from 4 cases, with most cases contributing four to six encounters that share a baseline shedding history. Encounters of the same case are therefore not independent, so encounter-level intervals overstate the effective sample size; the headline wake comparison is reported additionally with a case-clustered bootstrap and a case-level paired test (Appendix A). The partition is fixed once and used identically by every model and baseline.

For each frame the cache stores the mid-plane spanwise vorticity $\omega_z(192, 96)$, the span-averaged wall pressure $p_{\mathrm{wall}}(192)$, and the lift and drag coefficients C_L, C_D. The vorticity is normalised by a fixed pipeline: a spatial mask removes the solid and one adjacent cell layer to suppress the leading-edge finite-difference artefact, a per-encounter high-percentile clip caps spikes, and the field is scaled by the training standard deviation with no mean shift, preserving the vorticity antisymmetry. All representation and probe losses are computed in this normalised space; unnormalised values are used only for physical-units metrics and figures.

We evaluate forward closure against six physical observables that together span body-force, integrated-flow, and spatially distributed wake diagnostics. The lift and drag coefficients C_L and C_D are the body forces. The wall-normal impulse I_y is an integrated-flow quantity coupled to the wake-vortex dynamics. The wake enstrophy and the positive and negative circulation are spatially distributed wake observables: they integrate (signed) vorticity over the wake region and are the most demanding to forecast, because they are sensitive to the redistribution of vorticity that the LEV roll-up and shedding produce. The choice is deliberate: the wake observables can change sharply while the scalar forces change little, so they are the part of the encounter most likely to separate a representation that has captured the gust-vortex physics from one that has only captured its force signature. On these grounds we fix the wake enstrophy in advance as the single primary endpoint of the closure comparison; the other five observables are reported as secondary and descriptive. This pre-registration is what the family-wide multiplicity correction in Appendix A is applied against.

The six observables are defined on each frame of the masked vorticity field ω_z described above. The lift and drag coefficients C_L, C_D are as defined earlier in this section. On the wake window $\Omega_w = \{(x, y) : x/c \in [0.5, 4],\ |y/c| \le 1\}$ the wake enstrophy is

$$E_w(t) = \int_{\Omega_w} \omega_z^2 \,\mathrm{d}A, \tag{2.2}$$

and the signed wake circulations integrate the vorticity over its strongly rotational parts,

$$\Gamma^+(t) = \int_{\Omega_w,\ \omega_z > \omega_c} \omega_z \,\mathrm{d}A, \qquad \Gamma^-(t) = \int_{\Omega_w,\ \omega_z < -\omega_c} \omega_z \,\mathrm{d}A, \tag{2.3}$$

with threshold $\omega_c = 1$ in the cleaned vorticity units. The signed circulations at $\omega_c \in \{0.5, 1, 2\}$ are nearly collinear (pairwise Pearson well above 0.9 across held-out frames) and the predictive latent's circulation closure advantage over the reconstructive baseline is essentially unchanged across them, so the closure does not depend on this threshold. The wall-normal hydrodynamic impulse per unit span is

$$I_y(t) = \int_{\Omega} x\, \omega_z \,\mathrm{d}A \tag{2.4}$$

over the full cached field Ω. Every case uses the same fixed window and mask, and values are reported in the cache vorticity units integrated over chord-normalised area. The forecast horizon $H = 16$ is the sixteenth cached frame after impact, $t = 16\,\Delta t = 0.8\,c/u_\infty$.

Two definitional points bound how these observables are read. The impulse I_y is the mid-plane two-dimensional vorticity impulse, a diagnostic of wake-vortex transport, and not the impulse-theorem lift: the mid-plane slice omits the bound circulation, so on the simulation $\mathrm{d}I_y/\mathrm{d}t$ does not track the lift (Pearson correlation near zero on test_b, far from the value near -1 that a planar impulse-lift relation would give), and no closure result below should be read as recovering the impulse-theorem lift. More broadly, the wake and flow observables (the enstrophy, the two circulations, and I_y) are computed on the same

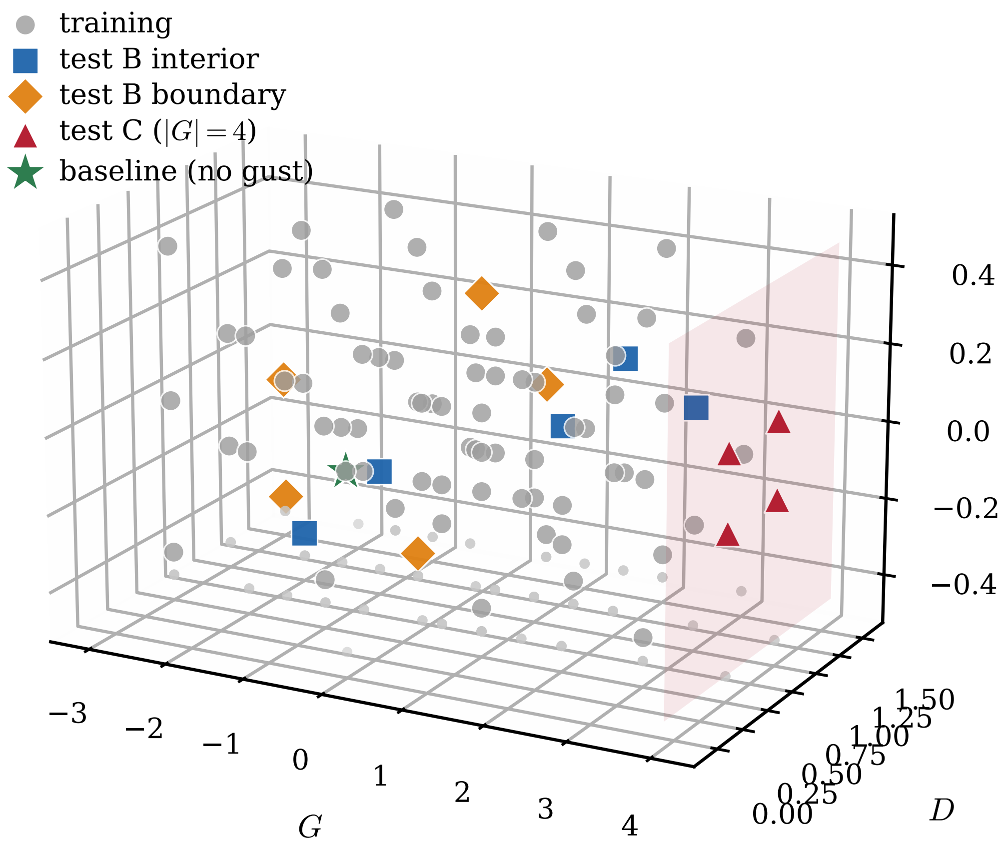


Figure 1. Parameter-space sampling of the 85 cases in the (G, D, Y) gust envelope, shown as one isometric view and coloured by split: training (229 encounters), the stratified in-distribution test set (test_b, 42 encounters, split into interior and boundary tiers), and the out-of-distribution extrapolation set (test_c, 24 encounters). The translucent plane marks the $|G| = 4$ extrapolation boundary and the grey points on the floor are the projection of every case, which shows that training mass concentrates near $Y/c = 0$, the reason the wall-normal offset is the least well resolved parameter (§4.6).

mid-plane slice the encoder consumes, so they omit the out-of-plane content quantified in §2.1 (roughly a fifth of the ω_z enstrophy in distribution), whereas C_L and C_D are true three-dimensional surface-integrated forces. The closure therefore compares mid-plane wake diagnostics against three-dimensional forces, and the mid-plane proxy is weakest at the $|G| = 4$ extrapolation boundary, where the out-of-plane fraction rises substantially.

3. Predictive latent representation and evaluation protocol

3.1. *Model and training objective*

The model takes no gust parameters: neither the encoder nor the predictor receives (G, D, Y). The predictive encoder E_θ maps the mid-plane vorticity field $\omega_z(192, 96)$ to a latent $\mathbf{z} \in \mathbb{R}^d$ at $d \in \{16, 32, 64\}$. It is a hybrid network: three convolutional downsampling stages produce a $24 \times 12 \times 256$ feature map, a six-layer vision transformer processes the

Table 2. Encoder and predictor configuration. The model is unconditional: no gust parameters $\mathbf{c} = (G, D, Y)$ enter the encoder or the predictor. Parameter counts are for the $d = 64$ configuration; two small auxiliary heads read observables off the encoder output during training (a lift head, 4×10^3 parameters, and an 80-dimensional wake-spectrum head, 3.5×10^4 parameters; §3.1).

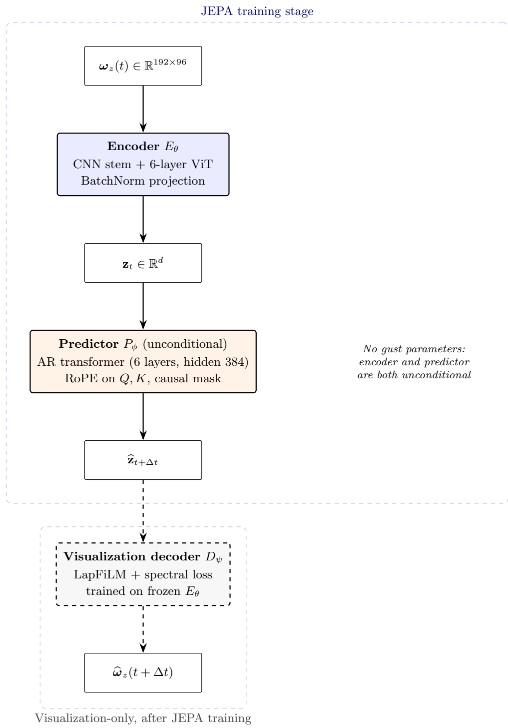
	Encoder E_θ	Predictor P_ϕ
role	field $\longrightarrow$ latent (unconditional)	latent dynamics (unconditional)
input	$\omega_z \in \mathbb{R}^{192\times 96}$	latent sequence $\mathbf{z}_{1:t}$
backbone	3 conv. stages + 6-layer ViT	6-layer autoregressive transformer
width / heads	256 / 8	384 / 16
tokens / context	$24 \times 12 = 288$ spatial	≤ 32 temporal (causal)
conditioning	none	none (RoPE temporal positions, causal mask)
latent read-out	class token $\longrightarrow$ linear $\longrightarrow$ BatchNorm	next-step latent
parameters	6.7M	16.2M

resulting tokens, and the class-token output is projected to $\mathbf{z}$ through a single linear layer with a batch-normalisation boundary, which the characteristic-function regulariser below requires. The encoder is a pure state map: it sees only the flow field, not the gust parameters $\mathbf{c} = (G, D, Y)$. This is the property that lets the latent itself serve as the observable a controller would have access to, since at deployment the gust parameters are not known a priori (§5).

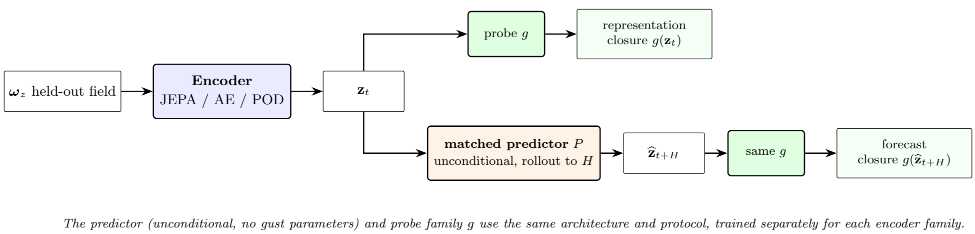
The predictor P_ϕ advances the latent in time. It is a six-layer autoregressive transformer with rotary positional embeddings and a causal mask, and it advances the latent from its own history alone, with no gust input, so the gust parameters enter nowhere in the model. Its internal structure is given in Appendix A. Figure 2(a) shows the encoder and predictor, and panel (b) sets out the matched-predictor evaluation protocol used throughout; the contrast between the predictive training signal and the reconstructive one, in which the reconstructive autoencoder passes the field at time t through an encoder and a decoder and incurs a field-space error whereas the predictive model encodes the fields at t and $t + \Delta t$ with shared weights and incurs a latent-space error between the predicted and observed target embeddings, with no field-space loss, is drawn in Appendix A (figure 14). Table 2 summarises the two networks.

The predictive encoder and predictor are trained jointly on a one-step latent-prediction term, a scheduled-sampling rollout term that exposes the predictor to its own errors, and an anti-collapse regulariser that prevents the encoder from collapsing to a constant. We use the characteristic-function regulariser of Balestriero & LeCun (2025); Maes *et al.* (2026), which matches the latent distribution to an isotropic Gaussian through a small number of random projections, with an automatic fall-back to a variance-covariance objective (Bardes *et al.* 2022) should the participation ratio of the latent indicate collapse. Its configuration is given in Appendix A.

Two auxiliary heads read aerodynamic observables off the encoder output during training: a lift head that maps $\mathbf{z}_t$ to $C_L(t)$, and a wake head that maps $\mathbf{z}_t$ to an 80-dimensional signed wake spectrum, both trained with small weights jointly with the representation and neither part of the latent-prediction loss. These heads are adopted from the observable-augmentation lineage (Fukami & Taira 2023, 2025). They are also, however, the reason the present predictive encoder is not a clean isolation of the training objective: the reconstructive baseline below carries a lift head but not a wake head, and the two encoders differ in architecture as well. We are explicit about this throughout. The headline comparison is therefore between encoder *families* as configured, and the



(a)



(b)

Figure 2. The predictive model and how it is evaluated. (a) The encoder maps the mid-plane vorticity field to a latent $\mathbf{z}_t$, and the autoregressive predictor advances the latent from its own history; no gust parameters enter the encoder or the predictor. The two auxiliary observable heads (lift and wake) read $\mathbf{z}_t$ during JEPA training with small weights and are not part of the latent-prediction loss; they are omitted from the schematic for clarity. The visualisation decoder is trained only after the encoder is frozen and is used solely for the diagnostic decodes of §4.6, never in the predictive loss. (b) Evaluation applies the same predictor and probe family g to every encoder family: a probe on the simulation-encoded latent measures representational closure, the headline, and the same probe on the rolled-out latent $\widehat{\mathbf{z}}_{t+H}$ measures forecast closure, reported as secondary.

controls of §4.2 are designed to separate the objective from the architecture and the wake supervision.

3.2. *Matched-predictor closure protocol*

The comparison rests on a single principle: every baseline is the same pipeline, and only the encoder differs. Each encoder produces a latent trajectory $\mathbf{z}_{1:T}$ from the vorticity fields, and an identical probe family maps a latent to a physical observable for evaluation (figure 2(b)). The probe family is held fixed across the three encoder families, and the predictor is shared by all of them. The latent dimension is matched within each family: JEPA at $d \in \{16, 32, 64\}$, the reconstructive autoencoder at $d \in \{3, 32, 64\}$ to include both the published $d = 3$ recipe and matched-capacity variants, and POD at $d \in \{16, 32, 64\}$. The predictive family is additionally swept to $d \in \{4, 8\}$ and crossed with a conditioned predictor variant in §4.2. This isolation is the only fair way to attribute representation quality to the encoder rather than to incidental differences in the readout. The reconstructive autoencoder uses its best-documented configuration, with ReLU activations, group normalisation, and a future-lift head at horizons $\{8, 16, 24\}$, rather than the strict published variant (tanh, no normalisation, a current-lift head), which we found gives a worse parametric probe; the comparison is therefore against a tuned baseline, not a hobbled one.

The headline figure of merit is representational closure: a fixed linear probe is applied to the latent encoded directly from a held-out field, and we ask whether it recovers each of the six observables, the wake in particular. This isolates what the encoder carries, with no rollout. We report it as the held-out coefficient of determination R^2 on test_b and test_c, with the per-observable mean absolute error alongside, and bootstrap 95% confidence intervals over encounters. As a secondary, confirmatory measure we also report forecast closure: the predictor is rolled recursively from the pre-impact context to $H = 16$, and the same probe maps the predicted latent to each observable. A small forecast error means the encoder produces a latent whose dynamics, propagated by the predictor, retain the information the probe needs; a large error means either that the encoder discarded that information or that the predictor lost it during rollout, a distinction the drift diagnostic below resolves. The representational closure is in table 4(a) and the forecast closure in table 4(b). We do not headline a forecast-R^2-versus-horizon curve; the rollout is reported to show that the latent stays usable, not as a validated forecast.

3.3. *Diagnostics, controls, and uncertainty*

Two diagnostics turn the closure result into a mechanism. The first is the latent drift, which addresses the fact that a predictor with low one-step error need not produce a useful long rollout, because each prediction is fed back as the next input and small geometric errors accumulate. We quantify how far a rollout has left the training manifold by the Mahalanobis distance of the rolled-out latent at horizon H, measured against the distribution of simulation-encoded latents for the same encoder, normalised by the Mahalanobis distance of the simulation-encoded latent itself. A ratio near one means the rollout stays within the natural spread of the training latents; a large ratio means it has gone out of distribution, so that the probe is then queried outside its support.

The second diagnostic is a parameter-only floor, which carries particular force here because the model never sees the gust parameters: it bounds what the parameters alone could explain, so that any closure above it is genuinely supplied by the latent state. We compute it with a kernel-ridge regressor that maps $\mathbf{c} = (G, D, Y)$ directly to each observable at $H = 16$, with no latent input, fit on training and evaluated on the held-out splits. On wake enstrophy this floor is low, so the latent's wake closure is not something the parameters could have supplied. At the impact frame the same parameter floor is high, because the just-released gust nearly determines the instantaneous state; the gap opens as the wake evolves (§4.1).

Because the predictive encoder differs from the reconstructive baseline in objective, architecture, and auxiliary supervision at once, the comparison is accompanied by controls that vary one factor at a time: a crossing of objective with architecture at matched auxiliary heads, an ablation of each auxiliary head, and a conditioned predictor variant that adds $\mathbf{c}$

Table 3. Encoder families compared under the matched-predictor protocol of §3.2. The model takes no gust parameters: none enter the encoder or the predictor. All three families share one autoregressive transformer predictor, training recipe, and probe family; only the encoder differs. "Aux. heads" lists the observable supervision attached to the encoder. The architecture and auxiliary-head columns differ across families, which is why the headline result is attributed to the predictive *family* and isolated only by the controls of §4.2.

Family	Objective	Encoder	Latent dim. d	Aux. heads
JEPA (this work)	predictive (latent)	CNN + ViT	16, 32, 64	lift + wake
Reconstructive AE	reconstructive (field)	CNN	3, 16, 32, 64	lift
POD	linear projection	none	16, 32, 64	none

Table 4. Held-out closure at $H = 16$ on the $n = 42$ test_b encounters, at matched $d = 64$, in two modes. *(a)* Representational closure, the headline: the coefficient of determination R^2 when a fixed linear probe is applied to the simulation-encoded latent (the latent encoded directly from the held-out field), larger is better. *(b)* Forecast closure, reported as confirmatory: the R^2 of the full-context rollout against the simulation observable (the variance baseline is the held-out split mean, so $R^2 < 0$ is worse than predicting that mean). The representation is where the families separate: the predictive latent renders the wake enstrophy linearly readable ($R^2 = 0.79$) where the reconstructive (Fukami) and linear (POD) latents are at or below zero. The forecast panel points the same way and is read as secondary; the per-encounter paired test is in table 8 and full probe values with bootstrap CIs in Appendix A.

(a) Representational closure: held-out R^2

	C_L	C_D	I_y	wake-enst.	circ. pos.	circ. neg.	mean
predictive (JEPA)	**0.86**	**0.69**	−0.21	**0.79**	**0.72**	**0.78**	**0.61**
reconstructive (Fukami)	0.56	0.23	0.13	−0.25	0.25	0.03	
linear (POD)	0.68	0.64	−0.69	−0.16	−0.15	0.32	

(b) Forecast closure (secondary): full-context rollout R^2

	C_L	C_D	I_y	wake-enst.	circ. pos.	circ. neg.
predictive (JEPA)	0.45	0.41	0.25	**0.43**	0.39	**0.74**
reconstructive (Fukami)	0.56	0.12	0.08	0.22	0.57	0.22
linear (POD)	**0.83**	**0.72**	−0.02	−0.60	−0.18	0.45

(§4.2). Every held-out R^2 and mean absolute error reported below carries three independent uncertainty signals, following the reporting protocol fixed with the data split: a 2000-resample bootstrap over the held-out encounters, the standard deviation across three encoder retrains from independent seeds, and a five-fold cross-validation over cases on the probe read-out. Their full specification is in Appendix A.

4. Results

4.1. *Representational wake closure and the parameter-only floor*

Table 3 summarises the three encoder families and the protocol. We distinguish two questions. Does the latent, encoded directly from a held-out field, render each observable readable by a fixed linear probe (representation quality, the headline, table 4(a))? And does the predictor, rolled out from the impact frame, track it (forecast quality, reported as secondary, table 4(b))? The first isolates the encoder; the second tests the encoder and the learned dynamics together. The result is established at the representation level, where the wake is the discriminator, and the rollout is shown as consistent confirmation rather than the load-bearing test.

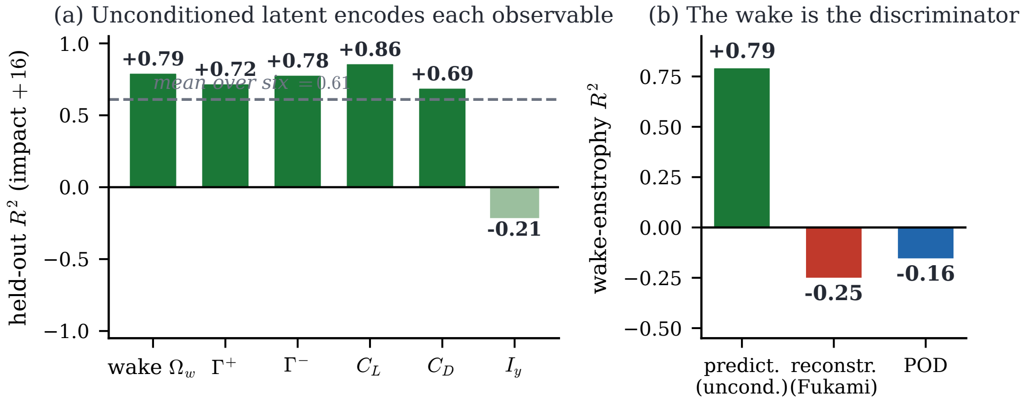


Figure 3. Representational closure of the predictive latent at $H = 16$ after gust impact, the headline result. (a) Held-out R^2 of a fixed linear probe applied to the simulation-encoded latent (no rollout) for each of the six observables, on the in-distribution test set (test_b, $n = 42$); the dashed line marks the mean over the six and the one weak observable, the mid-plane impulse I_y, is shaded light. (b) The wake-enstrophy R^2 across families: the predictive latent (0.79) against the reconstructive autoencoder (−0.25) and the linear basis (−0.16), each at $d = 64$. The wake is the discriminator, and only the predictive latent renders it linearly readable.

The wake is where the families separate. Probed directly from the held-out field sixteen frames after impact, the predictive latent renders the wake enstrophy linearly readable at $R^2 = 0.79$ (figure 3), while the reconstructive autoencoder and the linear basis are at −0.25 and −0.16, both below the predict-the-mean floor. A nonlinear (kernel-ridge) probe narrows this gap, recovering part of the wake from the matched-head reconstructive control (§4.2), so the claim is that the predictive objective renders the wake *linearly decodable*, not that the reconstructive encoding discards it irrecoverably. The forces and the circulations are read off the same latent cleanly, with C_L at $R^2 = 0.86$, the negative and positive circulations at 0.78 and 0.72, and the drag at 0.69; the mean over the six observables is 0.61. The one weak observable is the wall-normal impulse I_y ($R^2 = -0.21$), a mid-plane vorticity diagnostic that even the energy-optimal linear basis captures only modestly (§2.2), so its weakness is a property of the observable rather than of the predictive latent. The same ordering holds on the training, validation, and extrapolation (test_c) splits reported below.

The rollout points the same way and is read as confirmatory, not as the headline (table 4(b) and figure 4). A matched transformer predictor is trained on each family's latents under one protocol, and rolled out from the full pre-impact context; probing the wake enstrophy at $H = 16$, the forecast R^2 is 0.43 for the predictive latent, ahead of 0.22 for the reconstructive autoencoder and −0.60 for the linear basis. The forecast ordering is the same as the representational one but compressed, which is why we lead with the representation: the families separate most cleanly at the encoder, and the rollout confirms rather than carries the result. The forecast advantage is not statistically separable under case-level clustering (§4.3 and table 8), so we present it as consistent tracking, not as a forecast claim. On the extrapolation set (test_c, $G = +4$) the predictive wake-enstrophy forecast stays the strongest of the three while the integrated forces collapse, consistent with the three-dimensional observability boundary of §2.1.

The representational wake separation is the statistically robust core of the result. A per-encounter paired test (table 8), which cancels the encounter-to-encounter difficulty of the held-out set that every model shares, shows the predictive latent carrying the smaller wake-enstrophy error under direct encoding by a paired margin of 33.56 (case-clustered 95%

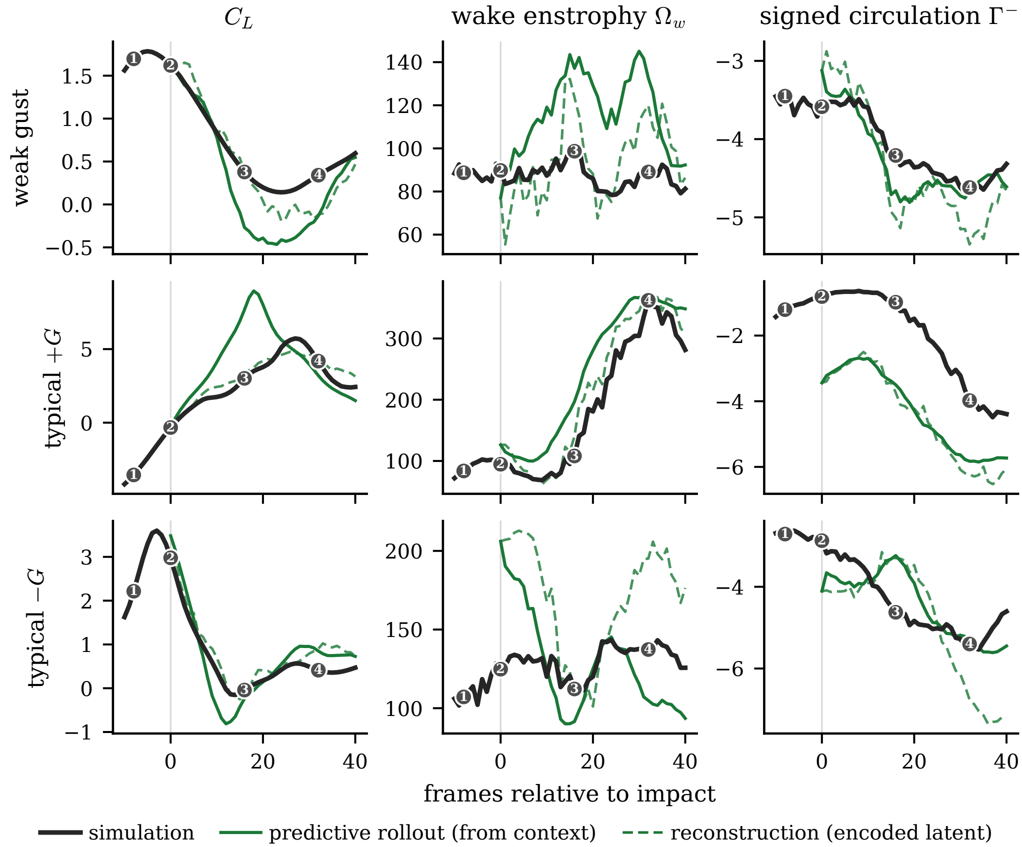


Figure 4. The predictive rollout tracks the encounter, shown qualitatively. Predicted observables through representative held-out encounters (rows), for the lift coefficient, the wake enstrophy, and the negative circulation (columns). The simulation is the bold reference; the rollout of the predictor, started from the pre-impact context, is the coloured line, and the dashed line is the reconstruction from the encoded latent. The rollout tracks the wake observables through the leading-edge-vortex peak and into recovery before drifting gently at long times. We report this as qualitative tracking, not as a closure R^2. Numbered glyphs mark the four encounter stages used throughout: baseline, impact, peak load, and recovery.

CI $[7.79, 58.72]$, excluding zero). We report the multiplicity correction at two levels and state honestly where the advantage stops: the representational wake-enstrophy advantage survives the pre-registered encounter-level family correction (one of 4 of twelve paired tests that survive Holm, together with the lift and the negative circulation), but it does not survive the stricter case-level correction, where the smaller number of held-out cases widens the interval. The forecast advantage points the same way but survives neither. All six observables are reported in both modes, not a selected subset, and the ordering is not uniform: the predictive latent leads on the spatially distributed wake observables (enstrophy and negative circulation), which is precisely the paper's claim. This separation is confirmed by the mechanistic evidence of §4.3 (the drift departure spectrum and the no-gust limit-cycle topology), where the predictive and reconstructive families separate consistently.

The parametric floor (table 5) confirms that the wake closure is not something the gust parameters supply on their own. Mapping $\mathbf{c} = (G, D, Y)$ to each observable at $H = 16$ with cross-validated regressors, the floor on wake enstrophy is negative ($R^2 = -0.18$ for

Table 5. Parametric (model-free) floor at $H = 16$ after impact. A cross-validated ridge regressor maps the gust parameters $\mathbf{c} = (G, D, Y)$ directly to each observable, with no latent input, fit on the training cases and read out on the in-distribution test split under the closure protocol ($R^2 = 1 - \mathrm{SSE}/\mathrm{SST}$ about the held-out mean). This is the model-free baseline: the predictive latent beating it (last column, the representational closure of table 4(a)) shows the representation carries flow state the parameters alone do not supply. On wake enstrophy the floor is negative ($R^2 = -0.18$): the gust parameters alone do worse than the held-out mean sixteen frames after impact, because the wake is then governed by the unfolding dynamics rather than by the release condition, whereas the predictive latent reads it at 0.79. On the forces, which track the gust forcing more directly, the floor is high.

Observable	parametric floor (test_b R^2)	predictive repr. (test_b R^2)
C_L	0.61	0.86
C_D	0.54	0.69
I_y	0.25	-0.21
wake-enstrophy	-0.18	0.79
circ. pos.	-0.05	0.72
circ. neg.	0.23	0.78

the linear regressor, -0.12 for the nonlinear one): the parameters alone do worse than predicting the held-out mean, while the predictive latent reads the wake at 0.79, clearing the floor by roughly 0.9. The floor is by construction strong on the observables that track the gust forcing directly, the lift (0.61) and the negative circulation (0.23): the predictive advantage is specific to the spatially distributed wake, not uniform across observables, and this is the paper's central distinction between the integrated forces, which the release condition largely fixes, and the wake, which it does not. The wake claim therefore rests on the representational closure (table 4(a)), the per-encounter paired test (table 8), and the drift mechanism (§4.3), with the parametric floor establishing that the gust parameters do not supply the wake state the latent encodes.

4.2. *Objective, architecture, and auxiliary-head controls*

The representational result of §4.1 is established, but the predictive latent *family* as configured renders the wake readable for reasons we have not yet separated: the predictive encoder differs from the reconstructive baseline not only in its training objective but also in its CNN+ViT architecture and its 80-dimensional wake supervision. Because the architecture and the wake supervision are confounds, we isolate the cause here, before turning to why the rollout stays on the manifold (§4.3). The controls in table 6 are the decisive crossing, run directly on the same models the paper reports and scored at the representational endpoint: a 2×2 crossing of objective {predictive, reconstructive} with architecture {CNN, CNN+ViT} at matched $d = 64$, with the auxiliary heads matched across all four cells, together with a predictive variant from which the wake head is removed. The decisive question is whether the predictive objective renders the wake readable more than the reconstructive one once the architecture and the wake supervision are held in common.

The controls (table 6) give a two-part answer, and we state both. First, the predictive objective renders the wake readable at *both* architectures with the auxiliary heads matched: the representational wake-enstrophy R^2 is 0.79 for the predictive CNN+ViT and 0.79 for the predictive CNN, while both reconstructive cells stay low (0.13 at CNN+ViT, 0.35 at CNN). The objective, not the architecture, carries the advantage: the two predictive columns do not separate, so the ViT is not the driver. Second, and equally important, the wake-observable supervision is necessary: the predictive model with the wake head removed collapses to a representational wake R^2 of 0.11, at the floor and below every other predictive cell. The result therefore requires the predictive objective *and* the wake

Table 6. Objective × architecture controls (decisive experiment), scored at the representational endpoint on the in-distribution test split. Held-out representational wake-enstrophy R^2 at $H = 16$ (test_b, fixed linear probe on the simulation-encoded latent, seed mean) for the four cells crossing {predictive, reconstructive} with {CNN, CNN+ViT} at $d = 64$, with the auxiliary heads (the instantaneous lift plus the 80-dimensional wake head) matched across all four cells, plus a predictive variant with the wake head removed. Predictive renders the wake readable at both architectures, while both reconstructive cells stay low; removing the wake head from the predictive model collapses wake readability to the floor, so the wake supervision is also necessary. The CNN and CNN+ViT predictive columns do not separate, so the ViT is not the driver.

Objective	Encoder	Aux. heads	repr. wake-enst. R^2
predictive	CNN+ViT	lift + wake	**0.79**
predictive	CNN	lift + wake	0.79
reconstructive	CNN+ViT	lift + wake	0.13
reconstructive	CNN	lift + wake	0.35
predictive	CNN+ViT	lift only	0.11

supervision together; it is not a property of the objective in isolation, nor of the architecture, nor of the wake head attached to an arbitrary objective (the reconstructive cells carry the same wake head and do not reach the predictive readability). We claim, accordingly, that the predictive objective renders the wake linearly readable over the reconstructive objective at matched architecture and matched supervision, with the auxiliary wake head a necessary component of the configuration that achieves it.

The latent dimension is the remaining control. The representational wake closure holds on a plateau from $d = 64$ down to $d = 16$ ($R^2 = 0.79$, 0.60, and 0.55 at $d = 64$, 32, and 16) and falls off below, so the reported $d = 64$ sits on the plateau and the wake is not recoverable at the smallest dimensions on this dataset. The latent code is correspondingly distributed rather than concentrated in a few coordinates, which §4.5 makes quantitative.

4.3. *Why the latent stays on the manifold: drift and topology*

The controls attribute the wake advantage to the predictive objective trained with wake supervision, and the representation is where that advantage is established. A separate question is whether the latent is usable under rollout, since a planner queries a forward model at the states its own rollout reaches. The answer is geometric: the predictive latent stays on its encoded manifold while the reconstructive one leaves it, which we establish with two coordinate-free diagnostics, the manifold departure of the rollout and the topology of the encoded encounter.

The predictive rollout stays on the encoded manifold. Two drift measures order the families oppositely, and the reconciliation is the mechanism (table 7). By relative Euclidean deviation from the encoded trajectory the reconstructive rollout is the *least* drifting (0.32 against 0.54 for the predictive at $H = 16$), yet by Mahalanobis distance to the training distribution it is an order of magnitude out (9.00 against 0.70 for the predictive, which stays in distribution). Both are true because the reconstructive rollout departs along directions of negligible encoded variance: a fraction 1.00 of its Mahalanobis departure lies in the near-null directions of the encoded covariance, where a small Euclidean step is a large Mahalanobis one, whereas the predictive rollout wanders within the broad isotropised cloud its anti-collapse regulariser maintains. The departure spectrum, not the ratio magnitude, is the robust statement: the reconstructive rollout exits the manifold along the directions the decoder never constrained. Ratios below one (the predictive and linear rollouts) mean the rollout is more in-distribution than the encoded held-out latents, the predictor mean-reverting toward the training cloud. This is the property a forward model needs, and it is

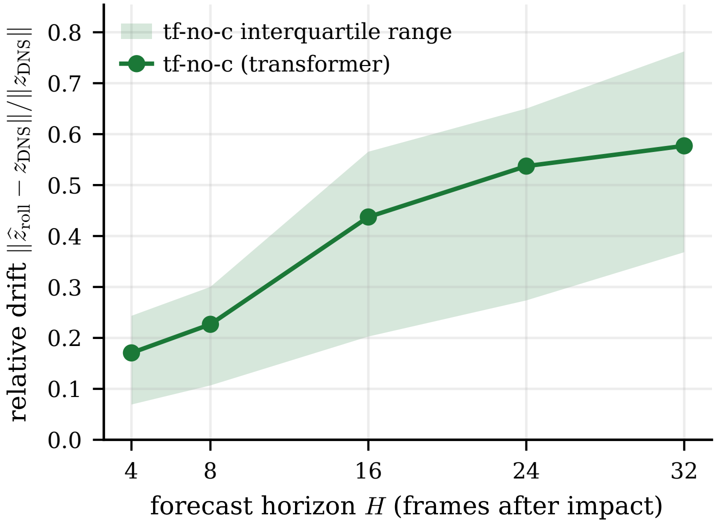


Figure 5. Relative drift of the predictive rollout on test_b versus forecast horizon H, measured as the deviation of the rolled latent from the simulation-encoded latent normalised by the encoded-latent scale, $\|\widehat{\mathbf{z}}_{\mathrm{rollout}} - \mathbf{z}_{\mathrm{DNS}}\| / \|\mathbf{z}_{\mathrm{DNS}}\|$. The transformer predictor (tf-no-c) drifts gradually and stays bounded. This relative deviation understates the family separation, which is the Mahalanobis manifold-departure of table 7: the reconstructive rollout exits along low-variance directions, small in this norm but large in Mahalanobis.

Table 7. Latent-drift diagnostic on test_b at $H = 16$, two measures. The relative Euclidean deviation of the rollout from the encoded trajectory and the Mahalanobis distance of the rolled-out latent to the training distribution order the families oppositely (Fukami least-drifting by the former, an order of magnitude out by the latter); the near-null fraction is the share of the Mahalanobis departure lying in the near-zero-variance directions of the encoded covariance, which reconciles the two. The Mahalanobis magnitude is reported at the minimally-regularised covariance used here; under a shrinkage estimator the ratios collapse toward one, so the robust statement is the departure direction (the near-null fraction), not the ratio size.

Family	rel. ℓ_2 drift	Mahalanobis dist.	near-null frac.
Fukami	0.32	**9.00**	1.00
JEPA (tf)	0.54	0.70	0.88
POD	0.84	0.84	—
β-VAE	0.53	0.87	—

why the qualitative rollout of figure 4 tracks the encounter through impact before drifting only at long times.

The departure spectrum points to a single geometric cause, established here two ways that do not depend on the latent coordinates: the Mahalanobis distance that shows whether the rollout leaves the training manifold, and the topology that shows whether the encoded encounter is a single cycle.

The mechanism behind the closure gap follows. A probe attached to a drifted rollout is queried outside the support it was fit on, so its output degrades for reasons unrelated to the predictor's one-step quality. A reconstruction objective places latent codes wherever the decoder finds convenient and does not constrain the geometry to be predictable in time, leaving the near-null directions the rollout then exits along; the predictive objective, regularised against collapse, produces an isotropised latent that supports stable rollout by construction. This is a representation property rather than a probe artefact: no probe in

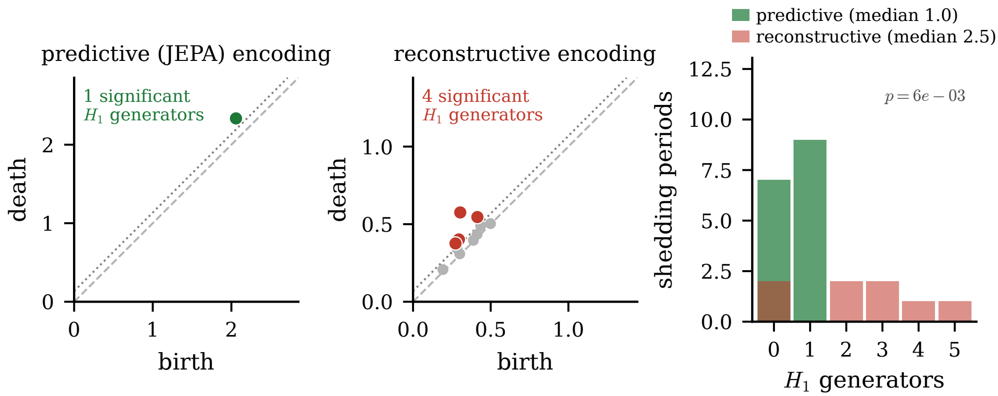


Figure 6. Persistent homology of the encoded no-gust limit cycle on test_b, under the per-family whitened metric that removes anisotropic coordinate scale. Left and centre: representative Vietoris–Rips H_1 persistence diagrams of the encoded predictive and reconstructive latents; coloured points are significant generators (lifetime above the dotted noise floor) and grey points are noise. Right: the significant-H_1 generator count over the undisturbed shedding periods. The predictive encoding is a single clean loop (median 1.0) while the reconstructive encoding fragments (median 2.5); the gusted fragmentation seen in raw coordinates does not survive the whitening and is not claimed.

the evaluated class recovers the wake from the reconstructive rollout once it has left the manifold the probe was fit on, while the same class reads it from the predictive rollout, so the reconstructive failure is the manifold departure and not an inaccurate predictor step.

The manifold departure is a rollout property; the second diagnostic asks whether the *encoded* encounter has the topology of a single cycle. The undisturbed shedding traces a limit cycle, and a gust encounter departs it, executes the LEV-formation and shedding excursion, and relaxes back toward it (Smith *et al.* 2024). We characterise the cycle with persistent homology, counting the significant one-dimensional generators (H_1) of a Vietoris–Rips filtration (Smith *et al.* 2024) on each encoded latent trajectory. Here honesty requires care with the metric. In raw latent coordinates the reconstructive encoding fragments a gusted encounter into several generators where the predictive encoding shows one, but the two encodings have very different per-coordinate scales by construction (the predictive latent is whitened toward isotropy by its anti-collapse regulariser, the reconstructive latent is not), and removing that anisotropic scale with a per-family whitening collapses the difference: under the whitened metric both families read a gusted encounter as essentially a single cycle (median 1.0 and 1.0 generators). The gusted-encounter fragmentation is therefore a property of the metric, not of the topology, and we do not claim it.

What survives the whitening is the cleanest control, the undisturbed no-gust limit cycle (figure 6). Even after per-family whitening the reconstructive encoding fragments the clean shedding cycle into several generators (median 2.5) while the predictive encoding keeps it as a single loop (median 1.0, as does the linear basis). The reconstructive encoding cannot represent even the undisturbed limit cycle coherently, which is an encoding property, distinct from the rollout manifold departure above and computed on the simulation-encoded latents.

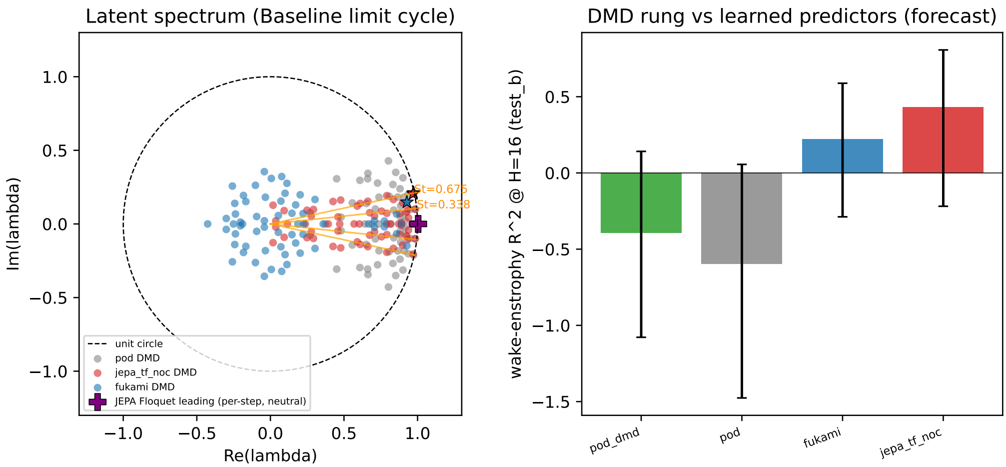


Figure 7. The latent dynamics carry the shedding spectrum but are not a linear operator. Left: dynamic-mode-decomposition eigenvalues of the encoded limit-cycle trajectory in the complex plane, by family, against the unit circle and the measured shedding-frequency ray; the predictive (JEPA) and linear (POD) latents place a marginally stable oscillatory pair at the shedding Strouhal number (0.66 and 0.68, moduli 0.991 and 0.996) while the reconstructive (Fukami) pair is damped and off-frequency (0.50). Right: the wake-enstrophy forecast skill of a best-fit linear operator on the linear basis (dynamic mode decomposition as a forecaster, −0.39) against the learned predictor (0.43), a margin of 0.82.

4.4. *Flow physics of the encounter and its latent image*

The closure and mechanism results are representational; this subsection reports what the analysis says about the flow itself, in measures that do not depend on any learned model, and where the predictive latent inherits those measures. Three results stand.

First, the latent dynamics carry the shedding spectrum. Fitting a linear operator to the encoded limit-cycle trajectory (dynamic mode decomposition) recovers a leading oscillatory eigenvalue at Strouhal number 0.66 in the predictive latent, against a measured shedding line at 0.68 in the undisturbed lift spectrum, with modulus 0.991, a marginally stable orbit; the linear basis recovers the same frequency (0.68) and the reconstructive latent is damped and off-frequency (0.50). Linearising the learned predictor about the orbit gives a leading Floquet multiplier of modulus 1.004, confirming the marginal stability intrinsically. The learned dynamics are not, however, a linear operator in disguise: rolling a best-fit linear operator on the linear-basis coordinates (dynamic mode decomposition as a forecaster) gives a wake-enstrophy forecast R^2 of −0.39, far below the learned predictor's 0.43 on the same observable, a margin of 0.82. The predictive latent therefore preserves the shedding spectrum that a linear basis also carries, while its nonlinear dynamics forecast the gusted wake that a linear operator cannot.

Second, the wake response scales with gust strength. The peak large-scale wake-enstrophy excursion grows monotonically with $|G|$, a trend that is robust to the scale band (Spearman 0.74 at $\sigma/c = 0.05$, 0.70 and 0.51 at the finer bands). The leading-edge vortex circulation budget shows a gust-sign asymmetry: negative gusts produce a larger and longer-lived leading-edge vortex than positive gusts at matched $|G|$, with a later circulation detachment (paired peak-circulation difference −0.240), quantifying the split-and-merge interaction the lineage describes qualitatively. The peak lift excursion and the peak vortex circulation are monotonically associated across the envelope (Spearman 0.75), reported as a correlation and not an impulse-theorem identity; the linear coefficient is weaker (Pearson 0.53, Appendix A).

Third, the gust resets the shedding phase. Within a case the encounters share gust parameters and differ only in release instant, so consecutive releases sample shifted shedding phases; the undisturbed cadence is reproduced to within a few hundredths of a radian, but any gust of $|G| \geq 0.25$ decorrelates the shedding phase between releases, and the mean phase step drifts monotonically with gust strength. The encounter does not preserve the pre-impact clock, it resets it, which is why a fixed-cadence release campaign undersamples the impact phase (Appendix A); we report the phase response within that measured coverage and treat the impact phase as a sampled rather than a controlled variable.

4.5. *The latent code is distributed*

The capacity ladder of §4.2 already indicated that the wake is not held in a few latent coordinates. Two measures make this quantitative. The wake forecast is a collective code: the gap between the wake readability of the full latent and that of its single most informative coordinate is 0.36 for the predictive latent, against 0.05 and 0.04 for the reconstructive and linear baselines, so no single predictive coordinate carries the wake and the information is spread across many. The energy-information curve agrees: the predictive latent needs about 32 leading principal components to reach its wake skill, where the reconstructive latent saturates at 12 and the linear basis at 1. The distributed code is the latent-space signature of the anti-collapse regulariser, which isotropises the representation, and it is consistent with the rollout staying within the broad cloud of §4.3.

We do not claim that this code is spatially localised on identifiable structures. A gradient-saliency footprint of the wake-forecast direction through the frozen encoder overlaps the high-vorticity cores and the Q-criterion vortices *below* a permutation chance baseline (overlap 0.070 and 0.010 against the chance band), so the wake-forecast direction does not read the input preferentially at the leading-edge vortex or the shear layer. The wake code is distributed in the latent and not localised in the field, and we report it as such.

4.6. *Where the wake is kept and where it is lost in physical space*

The mechanism explains why the predictive rollout stays on the manifold; we now ask where in the gust envelope, and where in the flow field, the latent renders the wake readable. The parameter dependence reveals one clear weakness, and it is worth noting that the latent encodes these parameters despite never being trained on them. Probing the predictive latent for the gust parameters on test_b recovers the gust strength G and the core diameter D ($R^2 = 0.83$ and 0.65) but does not resolve the wall-normal offset Y ($R^2 = -0.03$, the confidence interval straddling zero). This is a property of the data rather than of the architecture: the training mass concentrates near $Y \approx 0$, so encounters at different offsets within the envelope are weakly separated along Y, and the latent inherits that weakness. Any deployment-time parameter estimate should treat Y as a nuisance variable the encoder does not resolve.

The same latent can also be read relative to the undisturbed shedding cycle. Because the undisturbed wake is a limit cycle (§4.4), an encounter is naturally described by its phase and amplitude relative to that cycle, the same reduction used to design gust-mitigation control for these flows (Fukami *et al.* 2024). The predictive latent of the undisturbed case traces a closed periodic orbit (figure 8): concatenating the no-gust episodes, the return distance is 1.0% of the orbit diameter and the episodes overlap the same loop, with a shedding period near the full-cycle Strouhal number 0.34 (§4.4), and a phase advances monotonically along it via the Hilbert analytic signal of the two leading principal components. This is the same object the spectrum of §4.4 identifies as a marginally stable orbit and the topology of §4.3 identifies as a single persistent cycle. A gust encounter departs this orbit and only partially relaxes back: the simulation trajectories do not fully return within the 120-frame window,

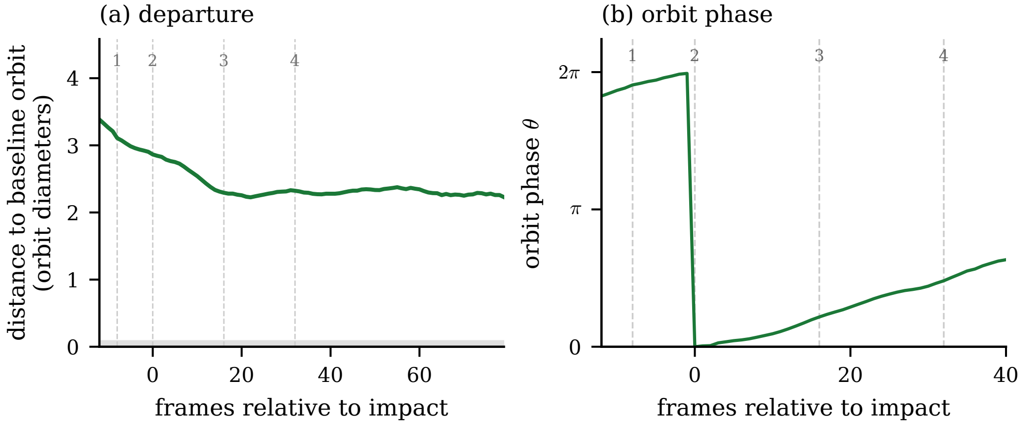


Figure 8. The gust encounter in the predictive latent. (a) Distance of the gust trajectory to the baseline orbit (full latent, in units of the orbit diameter; the grey band is the baseline orbit thickness): the encounter departs the baseline cycle and only partially relaxes back within the inter-gust window; the single-cycle topology of the encoded latent is established coordinate-free in figure 6. (b) The orbit phase relative to the impact frame. The numbered stages are those of figure 4.

which is the gust-release period itself ($6\,t/c$ at $\Delta t = 0.05\,t/c$, impact near $2\,t/c$), so only about $4\,t/c$ of post-impact relaxation is observed before the next release, a window-length limitation of the fixed gust cadence. We therefore read the latent cycle as the phase-amplitude coordinate a controller would use, and do not draw a family comparison from the incompletely-observed return.

We can also locate where in the envelope the predictive advantage concentrates. Rather than a per-stratum closure table, which the present test set is too small to populate, we read the per-encounter paired improvement $\Delta e = e_{\mathrm{AE}} - e_{\mathrm{JEPA}}$ in held-out wake-enstrophy error directly against each gust axis with a locally weighted regression and a bootstrapped band (figure 9). The advantage is not uniform across the envelope. It grows with the gust core diameter D (Spearman $\rho = 0.36$ on the encoded latent, significant), while its dependence on the gust strength G and the wall-normal offset Y is weak and not significant ($\rho = -0.13$ and $+0.03$). The predictive advantage therefore concentrates at the larger cores that most reorganise the wake; we do not read a Y trend, in keeping with the weak Y resolution of the probe above. The shedding phase at impact, the axis the gust-response literature emphasises (Fukami *et al.* 2024), is only weakly sampled by this dataset: impact is fixed near a single convective instant while the gust recurs about every two shedding periods, so the encounters fall into essentially two phase clusters and a phase-resolved trend is not identifiable here, which a design that varies the impact phase would isolate.

These parameter and phase readings are statements about numbers; the encounter is itself a statement about vortices. Decoded and reconstructed vorticity fields on held-out encounters (figure 10) connect the two, with one methodological caution and one new metric. Two scoping points apply throughout this subsection. The fields are encode-decode reconstructions, the visualisation decoder applied to the simulation-encoded latent, so the quantitative comparisons report the decoder's ceiling on what each latent carries from a faithfully encoded field, not a forecast rollout, whose latent leaves the data manifold for the reconstructive baseline (§4.3). And every quantitative physical-space claim is made on the validated large-scale band ($\sigma/c = 0.05$); the visualisation decoder is blurry at small scale by construction, so the large-scale leading-edge vortex and shear layer are tracked meaningfully while full-resolution and small-scale field fidelity are not claimed.

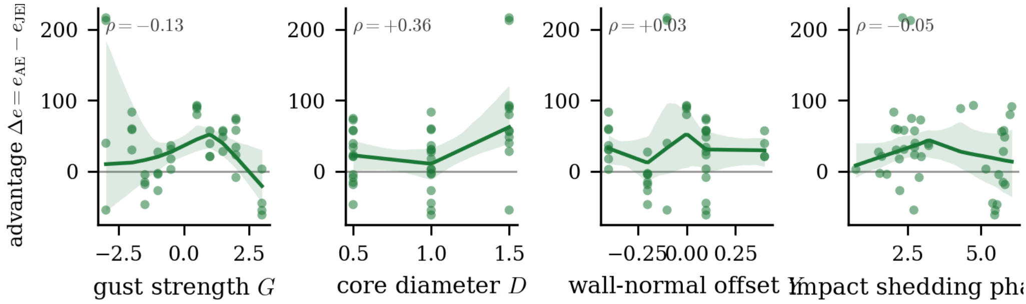


Figure 9. Per-encounter paired improvement $\Delta e = e_{\mathrm{AE}} - e_{\mathrm{JEPA}}$ in held-out wake-enstrophy error against the gust strength G, core diameter D, wall-normal offset Y, and the baseline shedding phase at impact ϕ (test_b, $n = 42$); positive Δe means the predictive latent carries the smaller error. Each panel shows the per-encounter values, a locally weighted (LOWESS) trend, and a bootstrap band over encounters. The predictive advantage grows with D (significant) but shows no significant G or Y trend; the shedding-phase axis falls into two clusters because impact is fixed near a single convective instant.

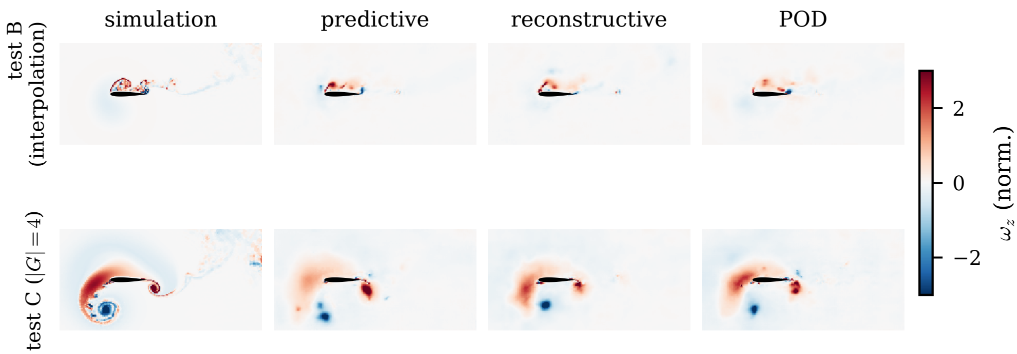


Figure 10. Decoded vorticity at the impact frame on held-out encounters (rows: an interpolation test_b encounter and a $|G| = 4$ test_c encounter) for the simulation and the predictive, reconstructive, and POD decodes (columns). The predictive decoder is blurry by design but localises the impingement; the reconstructive autoencoder collapses toward a near-uniform field on the held-out interpolation condition, while the linear basis is amplitude-accurate. All families degrade in the $|G| = 4$ regime, reported as the observability boundary of §2.1 rather than as a generalisation claim. Field-space similarity is read only as a labelled decode ceiling (§4.6).

The caution is that field-space similarity scores are misleading here. On held-out interpolation encounters the reconstructive autoencoder can attain a high structural similarity while collapsing to a near-uniform mean field that has lost the vortex-impingement signature, the high similarity merely reflecting bulk agreement with a field that is mostly quiescent away from the impact region. The predictive decode localises the impingement at the right position and sign but is blurry by construction, and the linear basis is amplitude-accurate since its objective is amplitude-optimal. Field-space similarity therefore cannot rank the latents, which is why we read it as a decode ceiling below.

This is why field-space scores cannot rank the latents here, and we are explicit about it. Decoding each family's faithfully encoded latent to a field and scoring the large-scale similarity against the simulation gives a decode *ceiling* (the best each decoder can do from a true latent): on test_b that ceiling is highest for the linear basis (0.71), then the predictive decode (0.62), then the reconstructive autoencoder (0.54). The linear basis leading this ranking is a decoder property, not a state-content one: its linear decoder reconstructs a smooth large-scale field well regardless of what the latent encodes, exactly the confound that the collapsed-reconstruction example above illustrates. Field decode fidelity is therefore reported as a labelled ceiling, with the explicit caveat that it conflates decoder reconstruction quality with latent state content; the latent comparison rests on the representational closure of §4.1, which has no decoder in the loop, not on these field scores.

Finally, the wake observables tie to the structures that carry the load. A Gaussian scale decomposition (large scale at filter width $\sigma/c = 0.05$, following (Odaka *et al.* 2026; Motoori & Goto 2019)) separates the large-scale vortical structures responsible for the lift peaks from the fine scale, applied to the simulation field and to each family's reconstruction (figure 11). On test_b the large-scale wake enstrophy of the predictive decode tracks the simulation better than the reconstructive autoencoder, on both correlation (Pearson 0.54 against 0.41 at $H = 16$) and amplitude (relative error 0.12 against 0.71): the reconstructive decode smooths the leading-edge vortex and shear layer away. The linear basis tracks the correlation comparably (0.58), the same decoder confound as the decode ceiling above, its linear decoder reconstructing the smooth large-scale field well, so the specific claim is that the predictive decode tracks the large-scale wake better than the reconstructive autoencoder, while field-space decode correlation does not separate it from the linear basis. This is where the wake-enstrophy advantage of §4.1 becomes a statement about the leading-edge vortex and shear layer rather than about a scalar. We are deliberately careful about the converse question, whether a family's *decode* tracks the leading-edge-vortex circulation per frame, because that test is confounded by decoder quality rather than by what the latent carries. Under the matched post-hoc decoders, the fraction of held-out encounters whose decoded circulation tracks the simulation is comparable across families and modest (0.29 for the predictive decode, 0.27 for the reconstructive, and highest for the linear basis at 0.48, whose linear decoder reconstructs the large-scale field best regardless of state content). We therefore do not claim that the predictive decode tracks the leading-edge vortex better than the alternatives; the wake result rests on the representational closure and the scale-resolved enstrophy tracking above, not on per-family decode fidelity.

4.7. *Wall-pressure observability*

At deployment the vorticity field is not available; only the wall pressure is. The representational result has an exact deployment mirror, and it is a main result of the paper: we ask which representation is the most recoverable from a sparse set of K wall-pressure taps over a pre-impact window, comparing both the recovered latent *state* and the recovered flow *field* across families under one kernel-ridge estimator and a target-blind (qDEIM) sensor placement (the placement, the per-family field-recovery breakdown, the parameter recovery, and the out-of-distribution boundary are in Appendix B).

The predictive state is the most recoverable (figure 12a). With $K = 8$ taps the held-out test_b state R^2 is 0.78, against 0.66 for the reconstructive autoencoder, 0.51 for the variational baseline, and 0.34 for the linear basis; the lead is seed-robust (0.78 ± 0.09 over four predictive seeds against 0.66 ± 0.04). The energy-optimal POD coordinates, optimal for reconstruction, are the hardest of the three to read from the wall: the same predictive objective that renders the wake readable from the field produces a state that is also legible from the wall. The wall pressure also recovers the wall-normal offset Y at $R^2 = 0.51$, so the weak Y resolution of §4.6 is a property of the mid-plane vorticity observable and the offset sampling, not of the pressure signal.

The recovered *field* tells a more nuanced story that we report in full. The pressure-to-latent map followed by each family's decoder reconstructs the large-scale field with similarity 0.57 for the predictive latent, 0.58 for the linear basis, and 0.48 for the reconstructive (figure 12b); the predictive latent beats the reconstructive significantly but ties the linear basis. The tie is a decoder property: the decode-ceiling gap (the loss from pressure recovery

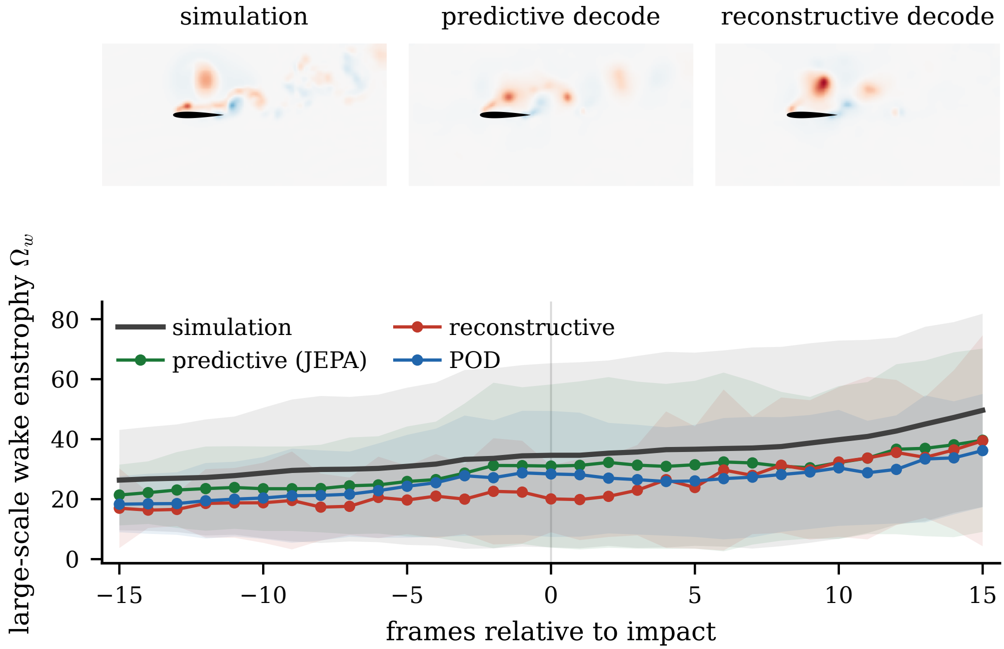


Figure 11. Gaussian scale decomposition ($\sigma/c = 0.05$) of the held-out reconstructions. Top: large-scale vorticity for the simulation, the predictive decode, and the reconstructive autoencoder at $H = 16$; the predictive latent retains the leading-edge vortex and shear layer the reconstructive one smooths away. Bottom: large-scale wake enstrophy through the staged encounter for the simulation, predictive, and reconstructive fields, on each split.

relative to decoding the true latent) is largest for POD (0.13 against 0.05 for the predictive), so the linear basis reaches field parity through its decoder's reconstruction ceiling, not through better state recovery, where it is the weakest of the three. The clean comparison is therefore the state recovery: the predictive latent is the most recoverable state from sparse wall pressure, and recovers the field at least as well as any matched representation.

5. Discussion

5.1. *Interpretation and scope*

The central finding is representational, and it concerns what the latent renders readable: the predictive latent makes the wake structure linearly decodable where the reconstructive and linear families do not. Under a shared probe family the predictive latent reads the held-out wake at $R^2 = 0.79$ while the baselines are at or below zero; the advantage is concentrated on the spatially distributed wake observables, survives the pre-registered encounter-level multiplicity correction (though not the stricter case-level one), and the forward rollout is consistent with the same ordering but shown qualitatively rather than as a validated forecast. We state the claim as linear decodability rather than exclusive possession because a nonlinear probe recovers part of the wake from the matched-head reconstructive control: the predictive objective renders the wake linearly readable, and the established reconstructive and linear recipes do not make it readable at all by the evaluated probe class. The mechanism is geometric. A reconstruction objective fixes the latent only up to a diffeomorphism (Kelshaw & Magri 2024), so the decoder is free to place codes anywhere it finds convenient and nothing constrains the latent geometry to be predictable in time; under autoregressive rollout the reconstructive latent leaves its

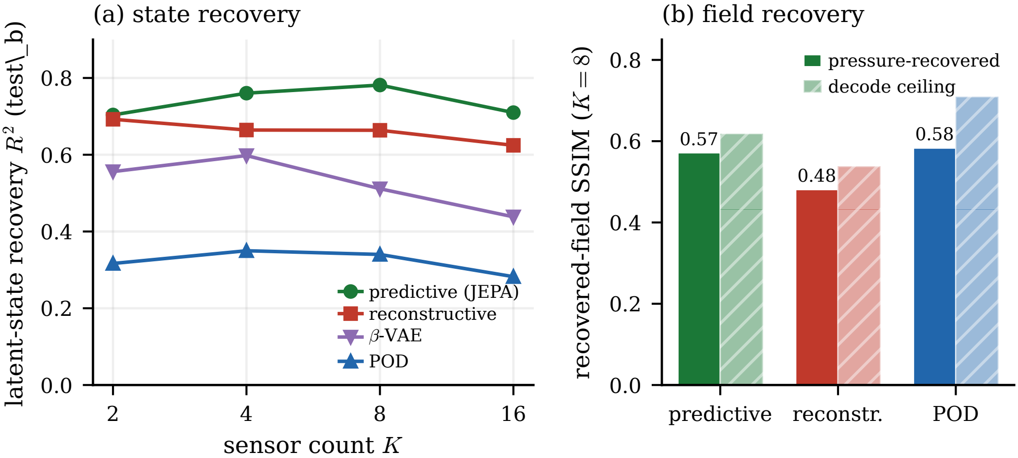


Figure 12. Cross-family pressure observability under the target-blind qDEIM sensor placement. (a) Latent-state recovery R^2 versus sensor count K: the predictive (JEPA) state is the most recoverable, the linear basis the least. (b) Recovered-field large-scale similarity by family (pressure to latent to decoder), with the decode-ceiling shown alongside: the predictive latent beats the reconstructive autoencoder and ties the linear basis, whose field parity is its decoder reconstruction ceiling rather than better state recovery.

own training manifold along the very directions the decoder never constrained, and a probe queried on that drifted state fails for reasons unrelated to the predictor's one-step accuracy. The predictive objective, regularised against collapse, instead organises the latent so that the encounter is a recurrent cycle that survives rollout, the on-manifold property that data-driven state-space and Koopman models of coherent-state dynamics on invariant manifolds rely on (Constante-Amores & Graham 2024), and that the encoded no-gust limit cycle exhibits even under a metric that removes coordinate scale. The manifold departure of the reconstructive rollout is the parametric-gust counterpart of the catastrophic-divergence tail reported for reconstruction-based latent dynamics in controlled wakes (Solera-Rico *et al.* 2025); the predictive objective is the common cure. The wake observables are the sharpest discriminator precisely because they can change while the scalar forces do not: the gust parameters alone fail to predict the wake at the readout horizon (the parametric floor of §4.1 is negative) yet predict the forces well, so a representation can carry the force signature of an encounter without carrying the vorticity redistribution that produces it, and that is exactly the reconstructive latent's failure mode.

The coordinate-level reading of §4.5 explains why the wake, and not the forces, separates the families. The wake forecast is a collective code in the predictive latent: the gap between the readability of the full latent and that of its single most informative coordinate is 0.36, against 0.05 and 0.04 for the reconstructive and linear baselines, whose best single coordinate is already as informative as the whole latent (figure 13). The forces, by contrast, are carried redundantly in every family. An energy or reconstruction objective therefore retains the force signature, present in many coordinates, but not the distributed combination that carries the wake, which is exactly the observable on which the predictive latent leads and which the gust parameters alone cannot supply (the parametric floor of §4.1 is negative on the wake). We read this as a description of correlations, not a causal decomposition: these are observational probes on the frozen encoders, alongside the information-theoretic modal analyses of related flows (Martínez-Sánchez *et al.* 2023; Arranz & Lozano-Durán 2024; Koshikawa *et al.* 2026).

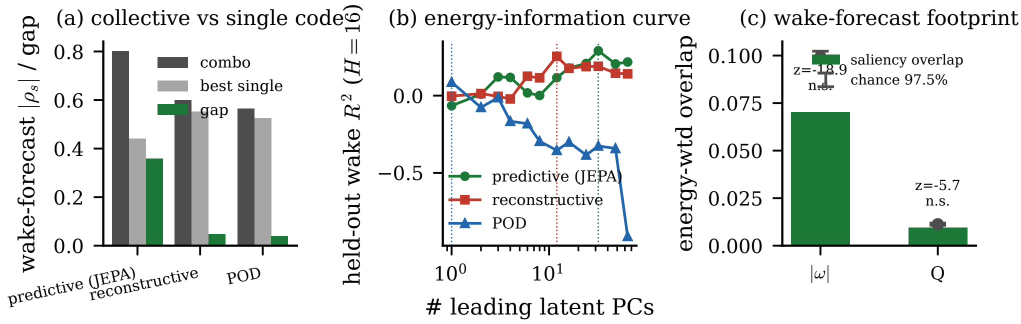


Figure 13. The wake is a distributed latent code, on test_b. Left: the gap between the held-out wake readability of the full latent and that of its single most informative coordinate, by family; the predictive (JEPA) latent shows a large gap (0.36) where the reconstructive (Fukami, 0.05) and linear (POD, 0.04) latents do not, so no single predictive coordinate carries the wake. Centre: the held-out wake skill versus the number of leading principal components, which knees at about 32 components for the predictive latent against 12 and 1 for the baselines. Right: the gradient-saliency footprint of the wake-forecast direction overlaps the high-vorticity cores and Q-criterion vortices below the permutation chance band, so the distributed code is not spatially localised. Observational probes on the frozen encoders.

The wake organisation is low-dimensional. The representational wake closure holds on a plateau as the latent dimension is reduced, from 0.79 at $d = 64$ to 0.60 at $d = 32$ and 0.55 at $d = 16$ (§4.2), so the representation declines gracefully rather than collapsing, and the qualitative ordering, predictive leading while the reconstructive and linear latents stay negative, is preserved throughout. The encoded latent is itself concentrated in a handful of directions: the $d = 64$ latent has a participation ratio of about 1.7, with the leading principal component carrying roughly three-quarters of the variance and six components spanning 90%, the dominant direction tracking the gust-strength dynamic range. The reduction in closure as the dimension falls is therefore a loss of precision on the wake observables, not of the geometric mechanism, which is consistent with the distributed code above: the wake is spread across many low-variance directions, and trimming them costs sharpness before it costs the ordering.

The mechanism above invites a strong reading, so we are deliberate about the scope of the claim. The predictive encoder differs from the reconstructive baseline in three respects at once: the training objective, the CNN+ViT architecture, and the wake supervision. The parameter-only floor removes the gust parameters as an explanation, all the more since the model never sees them, and the matched-architecture, matched-auxiliary 2×2 controls of §4.2 separate the remaining two. They give a two-part answer that we state in full. The predictive objective does keep the wake better than the reconstructive objective at both a CNN and a CNN+ViT encoder with the auxiliary heads matched, and the two architecture columns do not separate, so the gain is the objective and not the architecture. The wake supervision, however, is a necessary part of the configuration that achieves it: a predictive model with the wake head removed collapses below the predict-the-mean floor on wake closure. The claim is therefore that the predictive objective renders the wake readable *when trained with wake supervision*, not that the objective in isolation does, and not that the wake head attached to an arbitrary objective does, since the reconstructive cells carry the same head and do not reach the predictive readability. The linear basis, for its part, remains genuinely useful: it is the most competitive family on the integrated impulse and is amplitude-accurate in reconstruction, so the result is not that POD is obsolete but that it does not produce a latent from which the wake observables are linearly readable.

5.2. *Limitations and outlook*

Three limitations bound the claims. First, the wall-normal offset Y is weakly resolved because the training mass concentrates near $Y \approx 0$; this is a data limitation, and Y-conditional statements should be read with it in mind. Second, the $|G| = 4$ extrapolation degrades for a physical reason, not merely a statistical one. As set out in §2.1, the encoder observes a single mid-plane vorticity slice whose information no longer determines the three-dimensional $|G| = 4$ dynamics (Fukami *et al.* 2025), so test_c is the observability boundary of that representation, not a parametric-generalisation claim. Third, the visualisation decoder is a diagnostic on the frozen encoder, never part of the predictive loss; a reader expecting sharp reconstructions from a predictive latent will be disappointed, which is the explicit trade the objective makes. To these we add, as a fourth, the co-necessity of the wake supervision established in §5.1: the predictive objective renders the wake readable in the configuration tested, with the wake head as a necessary component, not as a property of the objective stripped of all auxiliary supervision.

These limitations come into sharpest focus against the use the architecture is ultimately meant for. The predictor is, by construction, the latent dynamics function a model-based controller would plan against, and three properties of the architecture are relevant to that use: the latent renders readable the wake the load transient is built from, the latent is two to three orders of magnitude cheaper to evolve than the full field, and the rollout stays on the training manifold, which is the property that determines whether long-horizon plans remain meaningful since a planner queries the predictor on states reached by its own rollout. The latent itself is the observable a controller has access to, with no gust parameters that would have to be estimated first, and adding an actuation channel is a change of input rather than of architecture; the phase-amplitude reading of §4.6 connects the predictor directly to the sensitivity-function control designed for these flows (Fukami *et al.* 2024). Consistent with treating the world-model view of §1 as motivation only, we caution against over-reading the predictor as an interventional world model. The predictor takes no gust parameters and is trained to forecast the observed latent trajectory, so it is a forward model of the encounters in the training envelope, not a counterfactual model of the effect of changing the gust: it makes no claim to predict how an encounter would differ under a parameter it never sees.

The contribution is narrow by design: a controlled comparison on a single parametric flow, anchored to the representational closure of the wake rather than to in-distribution reconstruction, with a drift mechanism that explains the gap and a set of controls (§4.2) that bound its interpretation. What the comparison establishes is bounded accordingly. The predictive advantage is robust on the representational wake closure and in the drift mechanism; it is partial on parameter extrapolation, where the wake encoding survives at $G = +4$ but the integrated observables do not (§4.1), and on the wall-normal offset Y, which is harder to resolve than G and D (§4.6); and it does not extend to the interventional reading or the closed-loop tolerances, both of which fail in the controls that tested them. We claim the representational wake result and its mechanism, not a validated forecast, a controller, or a counterfactual model. The natural next step is to apply the same matched-predictor protocol to other parametric flow datasets, to test how general the predictive-versus-reconstructive trade reported here actually is.

6. Conclusions

We asked which reduced-order state renders the wake that governs the transient load linearly readable in parametric vortex-gust airfoil interactions, and reached a representational answer with a geometric mechanism: a predictive objective regularised against collapse builds a latent from which the wake is linearly decodable and whose rollout stays on the encoded manifold, whereas a reconstruction objective smooths the wake away and its rollout drifts off the manifold along directions of negligible encoded variance. We tested this by comparing a predictive latent (JEPA, which takes no gust parameters), a reconstructive observable-augmented autoencoder, and a proper orthogonal decomposition basis as reduced states at Re = 5000, under a protocol in which a single probe family was held fixed so that only the encoder varied, judging each first by the representational closure of the wake. On held-out cases the predictive latent renders the wake linearly readable where the others do not: its representational wake-enstrophy R^2 is 0.79 against the reconstructive and linear baselines at or below zero, the per-encounter advantage survives the pre-registered encounter-level multiplicity correction (though not the stricter case-level one), and the gust parameters alone do not supply the wake state (the parametric floor is negative), while the linear basis remains most competitive on the integrated impulse. The same separation appears coordinate-free in the rollout drift and in the topology of the encoded no-gust limit cycle. The latent also carries the flow's own clocks: its spectrum recovers the shedding frequency as a marginally stable orbit, the wake response scales with gust strength, and the gust resets the shedding phase. Rolled forward, the predictor tracks the encounter, which we show but do not present as a validated forecast. Because the predictive encoder also differs in architecture and auxiliary supervision, matched controls attribute the advantage to the predictive objective trained with wake supervision. The same predictive state is the most recoverable of the three from sparse wall pressure, so it is observable as well as a faithful carrier of the wake, the property a downstream estimator or controller would exploit. We do not claim a validated forecast, a controller, or a counterfactual model. The remaining limitations are the weak resolution of the wall-normal offset, the three-dimensional observability boundary at $|G| = 4$, and a rollout error that still prevents a closed-loop control demonstration.

Funding. *[Authors to insert funding sources and grant numbers.]*

Declaration of interests. The authors report no conflict of interest.

Data availability statement. The reduced-order-model and evaluation code, the configuration files, the v2 data-split manifest, and the analysis outputs that regenerate the tables and figures are deposited at https://doi.org/10.xxxx/zenodo.PLACEHOLDER under the MIT License (DOI to be finalised on acceptance). The processed per-encounter cache, or a representative subset sufficient to rerun the heavier steps, is available from the corresponding author. The raw direct numerical simulations were computed with the SOD2D solver (Gasparino *et al.* 2024) and are owned by the simulation collaborators.

Author contributions. *[Authors to insert contributions, for example in CRediT format.]*

Use of generative-AI tools. Generative-AI assistance (Anthropic Claude, accessed through the Claude Code command-line interface) was used during 2026 to help with data analysis, figure generation, and manuscript drafting. All numerical results were computed by the authors' own code and verified against the underlying data, and the authors take full responsibility for the content of the article.

Appendix A. Architecture, anti-collapse regularisation, and uncertainty quantification

Architecture detail

The encoder is a hybrid convolutional stem followed by a small vision transformer. Three downsampling convolutional stages map the single-channel 192×96 vorticity field to a 24×12 feature map at 256 channels, that is 288 spatial tokens, which a six-layer pre-norm transformer of width 256 with eight heads processes; the latent is read from a prepended class token through a one-layer projection whose output passes a BatchNorm, the layer the characteristic-function regulariser below requires. The encoder is unconditional and carries no information about the gust parameters. The predictor is a six-layer autoregressive transformer of width 384 with sixteen heads and dropout 0.1, with a causal mask and rotary temporal positions. The predictor advances the latent from its own history alone, with no gust input, so the gust parameters $\mathbf{c} = (G, D, Y)$ enter nowhere in

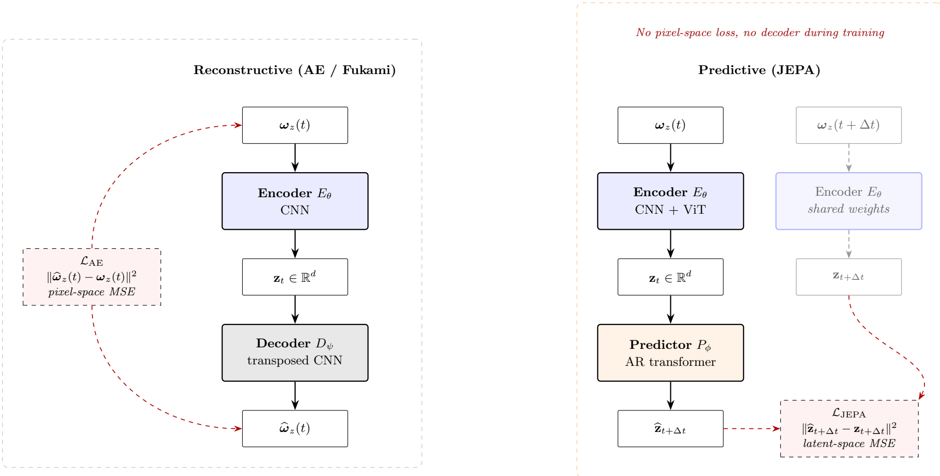


Figure 14. Predictive versus reconstructive training. Left: the reconstructive autoencoder encodes the field at t, decodes it, and incurs a field-space error. Right: the predictive model encodes the fields at t and $t + \Delta t$ with shared weights and incurs a latent-space error between the predicted and observed target embeddings, with no field-space loss and no decoder during training. This is the contrast the matched-predictor protocol of §3.2 evaluates; the visualisation decoder is trained only afterwards, on the frozen encoder.

the model. Training minimises the one-step latent prediction loss plus a scheduled-sampling rollout term and the regulariser below, with AdamW and bf16 mixed precision.

Anti-collapse regularisation

Because the encoder and predictor are trained end to end on latent-space prediction with no decoder, the latent is held away from a collapsed solution by an explicit regulariser rather than by a reconstruction term. We use SIGReg, the characteristic-function (Epps–Pulley) regulariser of the from-pixels joint-embedding line (Maes *et al.* 2026; Balestriero & LeCun 2025), with M random one-dimensional projections of the latent and a small weight (0.01) in the loss. As a safeguard the participation ratio of the latent is monitored every thousand iterations, with an automatic fall-back to a variance-covariance objective (Bardes *et al.* 2022) if it indicates collapse while the parameter probe is still uninformative; on the runs reported here the participation ratio stays healthy and the fall-back does not trigger.

Uncertainty quantification

Every held-out coefficient of determination and mean absolute error in the paper carries three independent uncertainty signals, following the reporting protocol fixed with the split. The first is a bootstrap confidence interval from 2000 resamples over the held-out test encounters, which is the dominant source of scatter at the present sample size (n = 42 for test_b, n = 24 for test_c) and is wide for the individual wake observables, as noted in §4.1. The second is the variance across three encoder retrains from independent random seeds, reported as the standard deviation in the objective-times-architecture controls of §4.2. The third is a five-fold cross-validation over cases on the read-out (probe) step, which guards against a probe that overfits the particular held-out split. Together these approximate what a full five-fold case-level cross-validation of the entire pipeline would buy, at a fraction of the training cost; a single split with bootstrap intervals is the community standard for this class of study.

Training-set fit

The predictive latent attains the highest in-distribution training fit of the forecast rollout among the families, but this is an in-distribution number, not the held-out result: the family ordering is established by the held-out representational closure of table 4(a), and the held-out forecast mean is markedly lower than the training fit, as it must be. We therefore do not tabulate the training-set fit, which would invite reading an in-distribution number as a result.

Table 8. Paired held-out closure, predictive ($d = 64$) against reconstructive ($d = 64$) on the $n = 42$ test_b encounters at $H = 16$, for the representational mode (probe on the simulation-encoded latent) and the forecast mode (full-context rollout). Δerr is the per-encounter mean of (reconstructive − predictive) absolute error in each observable's own units, so a positive value favours the predictive latent; the 95% interval is case-clustered (resampling the 10 cases), and the sign test counts the encounters on which the predictive error is the smaller of the pair. The wake enstrophy is the pre-registered primary endpoint (§4.1); the final column is the encounter-level sign-test p Holm-corrected across the twelve tests of the table (the pre-registered D165 family correction). Bold marks the four tests that survive that correction. We report this encounter-level correction as the headline and note the stricter case-level one in the text: the representational wake advantage survives the encounter-level Holm correction but not the stricter case-level one, where the smaller number of cases widens the interval. All six observables are shown, not a selected subset.

Mode	Observable	Δerr	case-clustered 95% CI	sign test	Holm p
repr.	C_L	**+0.552**	[+0.241, +0.949]	34/42, $p = 3.4\times10^{-5}$	**4.1×10^{-4}**
repr.	C_D	+0.134	[+0.00847, +0.309]	25/42, $p = 0.14$	0.7
repr.	I_y	−0.681	[−1.4, +0.0743]	17/42, $p = 0.92$	1
repr.	wake-enst.	**+33.6**	[+7.79, +58.7]	30/42, $p = 0.004$	**0.04**
repr.	circ. pos.	+0.398	[+0.0384, +0.912]	27/42, $p = 0.044$	0.35
repr.	circ. neg.	**+0.944**	[+0.302, +1.97]	33/42, $p = 1.4\times10^{-4}$	**0.0015**
forecast	C_L	+0.012	[−0.43, +0.424]	27/42, $p = 0.044$	0.35
forecast	C_D	+0.0813	[−0.0878, +0.251]	23/42, $p = 0.32$	1
forecast	I_y	+0.141	[−0.307, +0.61]	21/42, $p = 0.56$	1
forecast	wake-enst.	+16.8	[+1.06, +32.8]	26/42, $p = 0.082$	0.49
forecast	circ. pos.	−0.128	[−0.435, +0.194]	17/42, $p = 0.92$	1
forecast	circ. neg.	**+0.708**	[−0.0103, +1.72]	30/42, $p = 0.004$	**0.04**

Paired closure across all observables

Table 8 reports the per-encounter paired comparison of §4.1 for all six observables in both modes, the source of the wake-enstrophy significance cited in the main text. Pairing cancels the encounter-to-encounter difficulty that widens the marginal intervals of table 4.

Topological robustness

The significant-H_1 generator count of §4.3 declares a generator significant when its persistence exceeds a noise floor set at a fraction of the cloud diameter (the largest finite H_0 death), the canonical analysis using 5% of that diameter and all 120 trajectory frames. Following the convergence protocol of Smith *et al.* (2024) we sweep the floor over $\{2, 5, 10, 15, 20\}$% and the points per encounter over $\{120, 60, 40, 30\}$ (uniform subsampling), 20 settings in all. The predictive median is one generator in every setting. The predictive-versus-reconstructive Mann–Whitney separation is below 10^{-3} for floors from 2% to 15% at full (120-point) sampling and for floors up to 5% at half (60-point) sampling (the canonical 5%, 120-point value is 4.9×10^{-9}), while the reconstructive median falls from six generators at a 2% floor to one at a 20% floor as the aggressive floor filters the spurious generators. Under heavy subsampling to 30 points both medians approach one and the separation washes out. The robust statement is therefore the median-count separation over a defensible floor and sampling range, not the exact p; the case-clustered signed-rank test (10 of 10 cases at the canonical setting) holds across the same range. The full grid is in the released analysis package.

Return to the baseline orbit within the encounter window

The load-bearing caveat of §4.3, that the simulation gust trajectories do not fully return to the baseline orbit within the 120-frame encounter, is a consequence of the gust-release cadence rather than a strength-dependent failure to recover. Figure 15 holds the core diameter fixed at $D = 1.0$ with $G < 0$, averages the distance to the baseline orbit over each case's encounters, and sweeps $|G|$. The departure amplitude near impact is nearly independent of gust strength (peak distance 3.8 to 4.2 orbit diameters across $|G| \in [0.5, 3.0]$), every trajectory is still contracting toward the orbit at the end of the window, and the strongest gust ends closest rather than farthest. The residual offset at the window's end is therefore set by the $\sim 4\,t/c$ of post-impact relaxation the cadence allows and by the core diameter, a larger core leaving a more persistent wake, not by $|G|$. This window-truncated return is common to all three encoders and does not affect the relative drift comparison of §4.3, which contrasts the predictive and reconstructive rollouts over the same window.

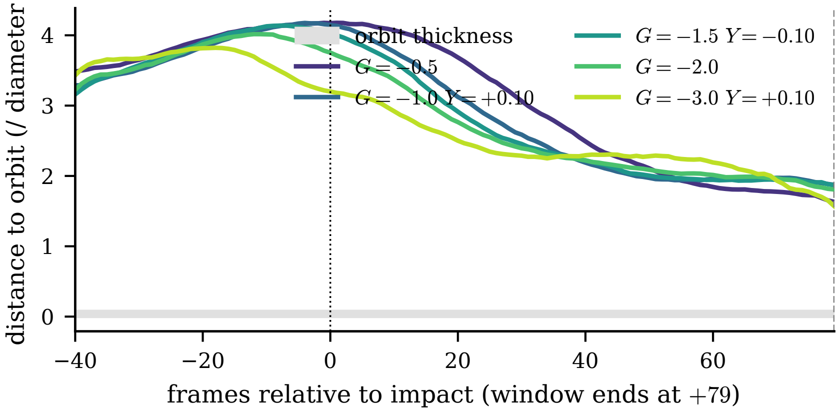


Figure 15. Diameter-controlled return to the baseline orbit. Distance of the encoded gust trajectory to the no-gust orbit (predictive latent, in orbit diameters; grey band is the orbit thickness), at fixed core diameter $D = 1.0$ and $G < 0$, averaged over each case's encounters, for five gust strengths. The departure amplitude is nearly independent of $|G|$ and all strengths are still contracting when the next gust is released at +79 frames, so the non-return is a window-length effect, not a strength-dependent one.

Appendix B. Sparse wall-pressure state estimation

At deployment the vorticity field is not available; only the wall pressure is. We ask how much of the latent state, the gust parameters, and the recovered flow field can be read from a sparse set of K wall-pressure taps over a pre-impact window, and through which representation. The input is the span-averaged pressure on the 192 surface taps; the pressure-to-latent map is a kernel-ridge regressor with a radial-basis kernel on the flattened K-sensor by pre-impact-window feature vector, fitted on the training encounters and applied unchanged on the held-out set. The window has $W = 30$ frames ending at the readout instant (impact plus $H = 16$, the closure-matrix primary endpoint), so the pressure is available up to but not beyond the readout. This is a feasibility comparison across representations, not a closed controller.

Target-blind placement

The placement we use throughout is qDEIM, the QR-pivot optimum of the wall-pressure modes (Manohar *et al.* 2018). It is target-blind and family-independent: it places taps to reconstruct the pressure field, with no knowledge of the latent it is later asked to recover, and the same eight taps are used for every representation. The cross-family recoverability ordering of §4.7 therefore cannot be an artefact of a bespoke latent-aware sensor selection. A target-aware greedy selection (target-conditioned sensor importance, which adds at each step the tap that most improves recovery of the leading latent component from the already-selected taps) is also available, and its only measurable advantage over qDEIM is in the scarce-sensor regime; by $K = 8$ the wall pressure is information-rich enough that the placement choice no longer separates the families, which is why we report the target-blind optimum.

State recovery and its anisotropy

With $K = 8$ taps the held-out test_b state R^2 orders the families as in the main text (§4.7): predictive 0.78, reconstructive autoencoder 0.66, variational baseline 0.51, linear basis 0.34. The variance-weighted R^2 is reported alongside the mean canonical correlation between the estimated and true latent, because the two expose different things. The linear basis has the highest canonical correlation (0.64 against 0.56 for the predictive) yet the lowest state R^2: the wall pressure aligns with many POD directions but those are largely the low-variance ones, so a high directional agreement recovers little of the state's energy. The predictive latent has the lower canonical correlation but recovers the high-variance directions that carry the state, which is the quantity a downstream probe reads.

Recovered field is a different axis from recovered state

Decoding the pressure-estimated latent through each family's frozen decoder gives a recovered flow field (figure 16). Scored by structural similarity, the predictive latent reaches 0.57, the linear basis 0.58, and the reconstructive 0.48: the predictive latent beats the reconstructive but ties the linear basis, and the tie is not significant (one-sided sign test $p = 0.860$). The tie is a decoder property, not a state-recovery one. The decode

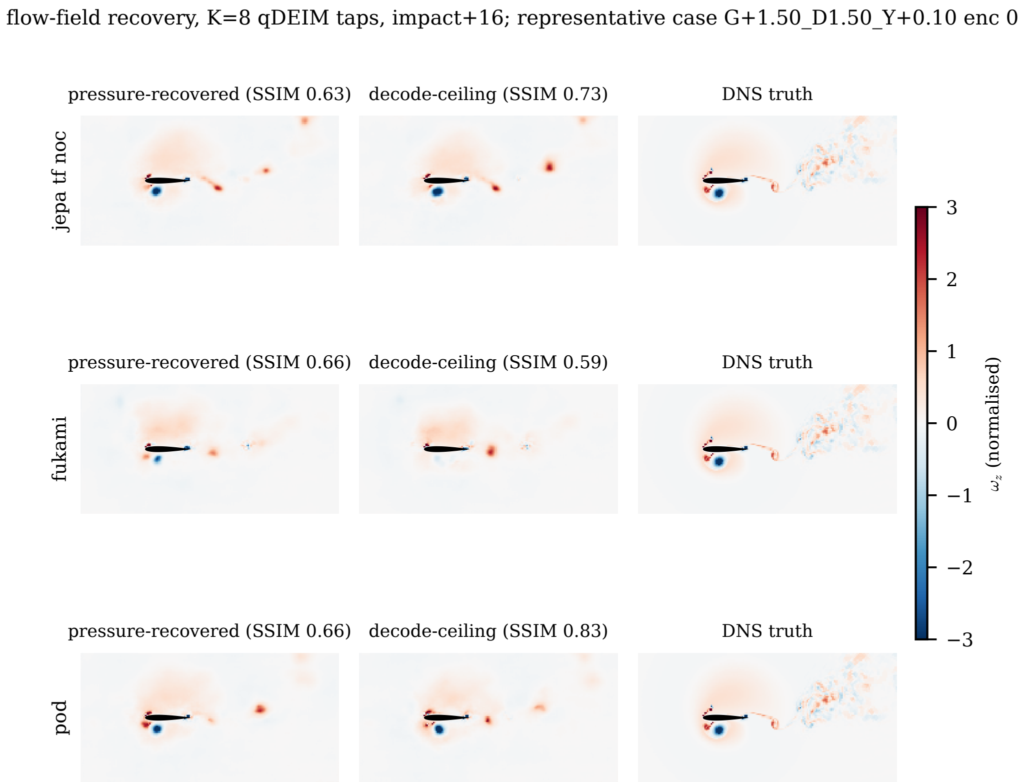


Figure 16. Flow recovered from $K = 8$ target-blind (qDEIM) wall-pressure taps on a representative held-out encounter (the typical-difficulty case $G = +1.50$, $D = 1.50$, $Y = +0.10$), one row per family. Columns: the field decoded from the pressure-estimated latent, the decode ceiling (the same decoder applied to the true encoded latent, with no pressure step), and the simulation. The per-panel similarity makes the appendix point visible: the linear basis reaches field parity through the highest decode ceiling, not through better pressure-to-state recovery, where the predictive latent has the smallest ceiling gap. The decoder is blurry by design (§4.6); the comparison is between the columns of the same row.

ceiling, the similarity of the field decoded from the *true* encoded latent with no pressure step, is highest for the linear basis (0.71 against 0.62 for the predictive and 0.54 for the reconstructive), so the linear basis reaches field parity through its decoder's reconstruction ceiling while having the weakest state recovery of the three. The pressure-to-state gap, the ceiling minus the pressure-recovered similarity, is correspondingly largest for the linear basis (0.13 against 0.05 for the predictive): the part of the field error attributable to imperfect pressure-to-state recovery is smallest for the predictive latent. The volume relative error and peak-vorticity retention agree with the similarity ordering at the impingement scale (predictive 0.92 relative error and 0.92 peak retention, against 0.78 for the linear basis). The state-recovery winner is therefore not the field-recovery winner (0 of the two orderings coincide), and the clean comparison is the state recovery, where the predictive latent is unambiguously the most recoverable.

Parameters and the out-of-distribution boundary

The gust parameters recover directly from the same eight taps, with the wall-normal offset the weakest (Y at $R^2 = 0.51$ on test_b), consistent with Y's weak resolution throughout the paper and confirming that the limit is the mid-plane vorticity observable and the offset sampling, not the pressure signal. The impact lift read through the latent ($R^2 = 0.43$) is no better than reading it straight off the pressure, in keeping with the main-text point that the wall reads the scalar lift well but the wake structure only through the representation. On the out-of-distribution test_c set ($|G| = 4$) every estimator and every family gives negative state R^2, the same three-dimensional observability boundary of §5.2 that bounds the rest of the model.

Robustness and a non-result

The frame-0 gust-release clip of the re-release encounters leaves the result unchanged: removing the clipped frame from the pressure window shifts the predictive state recovery by 0.001 in R^2, within noise. We do not claim wake recovery from the wall: a pressure-to-latent-to-frozen-wake-probe route reaches only $R^2 = 0.18$ for the predictive latent, a direct pressure-to-wake regression 0.17, and the predictive advantage over the best reconstructive family is not significant (one-sided sign test $p = 0.140$). The wake is linearly readable from the encoded field (§4.1) but not from this many wall taps over this window; the deployment comparison is the state and the field, not the wake.

References

Arranz, Gonzalo & Lozano-Durán, Adrián 2024 Informative and non-informative decomposition of turbulent flow fields. *Journal of Fluid Mechanics* **1000**, A95.

Assran, M., Duval, Q., Misra, I., Bojanowski, P., Vincent, P., Rabbat, M., LeCun, Y. & Ballas, N. 2023 Self-supervised learning from images with a joint-embedding predictive architecture. In *CVPR*.

Assran, M. & others 2025 V-JEPA 2: self-supervised video models. *arXiv preprint arXiv:2506.09985* .

Balestriero, R. & LeCun, Y. 2025 LeJEPA: provable and scalable self-supervised learning without the heuristics. *arXiv preprint arXiv:2511.08544* .

Bardes, A., Ponce, J. & LeCun, Y. 2022 VICReg: variance-invariance-covariance regularization for self-supervised learning. In *ICLR*.

Berkooz, Gal, Holmes, Philip & Lumley, John L. 1993 The proper orthogonal decomposition in the analysis of turbulent flows. *Annual Review of Fluid Mechanics* **25**, 539–575.

Brunton, Steven L., Noack, Bernd R. & Koumoutsakos, Petros 2020 Machine learning for fluid mechanics. *Annual Review of Fluid Mechanics* **52**, 477–508.

Constante-Amores, C. Ricardo & Graham, Michael D. 2024 Data-driven state-space and Koopman operator models of coherent state dynamics on invariant manifolds. *Journal of Fluid Mechanics* **984**, R9.

Cremades, Andrés, Martínez-Sánchez, Álvaro, Lozano-Durán, Adrián & Vinuesa, Ricardo 2026 X-CAL: Explaining latent causality in physical space for fluid mechanics. *arXiv preprint* ArXiv:2601.03311.

Eldredge, Jeff D. & Jones, Anya R. 2019 Leading-edge vortices: Mechanics and modeling. *Annual Review of Fluid Mechanics* **51**, 75–104.

Fukami, Kai & Araki, Ryo 2026 Information-theoretic machine learning for time-varying mode decomposition of separated aerodynamic flows. *AIAA Journal* **64** (2), 605–613.

Fukami, Kai, Nakao, Hiroya & Taira, Kunihiko 2024 Data-driven transient lift attenuation for extreme vortex gust–airfoil interactions. *J. Fluid Mech.* **992**, A17.

Fukami, Kai, Smith, Luke & Taira, Kunihiko 2025 Extreme vortex-gust airfoil interactions at Reynolds number 5000. *Phys. Rev. Fluids* **10**, 084703.

Fukami, Kai & Taira, Kunihiko 2023 Grasping extreme aerodynamics on a low-dimensional manifold. *Nat. Commun.* **14**, 6480.

Fukami, Kai & Taira, Kunihiko 2025 Observable-augmented manifold learning for multi-source turbulent flow data. *J. Fluid Mech.* **1010**, R4.

Gasparino, L., Spiga, F. & Lehmkuhl, O. 2024 SOD2D: A GPU-enabled spectral finite elements method for compressible scale-resolving simulations. *Computer Physics Communications* **297**, 109067.

Ha, David & Schmidhuber, Jürgen 2018 Recurrent world models facilitate policy evolution. In *Advances in Neural Information Processing Systems 31 (NeurIPS)*. ArXiv:1803.10122.

Jones, A. R., Cetiner, O. & Smith, M. J. 2022 Physics and modeling of large flow disturbances: discrete gust encounters for modern air vehicles. *Annu. Rev. Fluid Mech.* **54**, 469–493.

Kelshaw, Daniel & Magri, Luca 2024 Proper latent decomposition. *arXiv preprint arXiv:2412.00785* .

Koshikawa, Ryo, Araki, Ryo, Liu, Qiong & Fukami, Kai 2026 Convolutional causal learning for aerodynamic flows. *arXiv preprint* ArXiv:2601.19104.

LeCun, Yann 2022 A path towards autonomous machine intelligence. *Tech. Rep.*. OpenReview, version 0.9.2.

Lee, Kookjin & Carlberg, Kevin T. 2020 Model reduction of dynamical systems on nonlinear manifolds using deep convolutional autoencoders. *Journal of Computational Physics* **404**, 108973.

Linot, Alec J. & Graham, Michael D. 2023 Dynamics of a data-driven low-dimensional model of turbulent minimal Couette flow. *Journal of Fluid Mechanics* **973**, A42.

Liu, Zhecheng, Beckers, Diederik & Eldredge, Jeff D. 2025 Model-based reinforcement learning for control of strongly-disturbed unsteady aerodynamic flows. *AIAA Journal* ArXiv:2408.14685.

Lusch, Bethany, Kutz, J. Nathan & Brunton, Steven L. 2018 Deep learning for universal linear embeddings of nonlinear dynamics. *Nature Communications* **9**, 4950.

Maes, P.-A., Garrido, Q., Assran, M., Bardes, A., Rabbat, M., LeCun, Y. & Ballas, N. 2026 LeWM: latent world models for self-supervised video prediction. *arXiv preprint arXiv:2603.19312* .

Manohar, Krithika, Brunton, Bingni W., Kutz, J. Nathan & Brunton, Steven L. 2018 Data-driven sparse sensor placement for reconstruction: demonstrating the benefits of exploiting known patterns. *IEEE Control Systems Magazine* **38** (3), 63–86.

Martínez-Sánchez, Álvaro, López, Esteban, Le Clainche, Soledad, Lozano-Durán, Adrián, Srivastava, Ankit & Vinuesa, Ricardo 2023 Causality analysis of large-scale structures in the flow around a wall-mounted square cylinder. *Journal of Fluid Mechanics* **967**, A1.

Mohamed, Abdulghani, Marino, Matthew, Watkins, Simon, Jaworski, Justin & Jones, Anya 2023 Gusts encountered by flying vehicles in proximity to buildings. *Drones* **7** (1), 22.

Motoori, Y. & Goto, S. 2019 Generation mechanism of a hierarchy of vortices in a turbulent boundary layer. *J. Fluid Mech.* **865**, 1085–1109.

Murata, Takaaki, Fukami, Kai & Fukagata, Koji 2020 Nonlinear mode decomposition with convolutional neural networks for fluid dynamics. *Journal of Fluid Mechanics* **882**, A13.

Odaka, Hiroto, Lopez-Doriga, Barbara & Taira, Kunihiko 2026 Vortical similarities across laminar and turbulent extreme gust encounters. *J. Fluid Mech.* **1031**, R3.

Schmid, Peter J. 2010 Dynamic mode decomposition of numerical and experimental data. *Journal of Fluid Mechanics* **656**, 5–28.

Smith, Luke, Fukami, Kai, Sedky, Girguis, Jones, Anya & Taira, Kunihiko 2024 A cyclic perspective on transient gust encounters through the lens of persistent homology. *J. Fluid Mech.* **980**, A18.

Sobal, V., Zhang, W., Cho, K., Balestriero, R., Rudner, T. G. J. & LeCun, Y. 2025 Learning from reward-free offline data: a case for planning with latent dynamics models. *arXiv preprint arXiv:2502.14819* .

Solera-Rico, Alberto, García-Caspueñas, Patricia, Sanmiguel Vila, Carlos & Discetti, Stefano 2025 The balance between compactness and forecast accuracy of data-driven latent-space reduced-order models in controlled wake flows. Under review, Journal of Fluid Mechanics.

Solera-Rico, Alberto, Sanmiguel Vila, Carlos, Gómez-López, M., Wang, Y., Almashjary, A., Dawson, S. T. M. & Vinuesa, R. 2024 β-variational autoencoders and transformers for reduced-order modelling of fluid flows. *Nat. Commun.* **15**, 1361.

Taylor, G. I. 1918 On the dissipation of eddies. *Reports and Memoranda, Advisory Committee for Aeronautics* (598).

Tran, Jonathan, Yeh, Chi-An & Taira, Kunihiko 2026 Using optimal transport aligned latent embeddings for separated flow analysis. *J. Fluid Mech.* **1027**, A24.

Zamani Ashtiani, S. & Fukami, Kai 2025 Data-driven time-dependent bases for turbulent airfoil wake–extreme vortex gust interactions. *arXiv preprint* ArXiv:2512.09523.

Zhong, Y., Amiri-Margavi, A., Babaee, H. & Taira, Kunihiko 2025 Optimally time-dependent modes of vortex gust–airfoil interactions. *Journal of Fluid Mechanics* **1006**, A18.